

Chasing Your Own Tail

The Market Stability Impact of Leveraged and Inverse
Exchange-Traded Funds

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Abstract:

This thesis explores the market stability impact of the positive feedback mechanism of leveraged and inverse exchange-traded funds' (ETFs) rebalancing trades. We explore potential market stability effects by analysing intraday minute-by-minute data for underlying stocks in the OMXS30 and aggregate rebalancing flows for ETFs tracking the index. We find that a 1 % index move causes ETFs to originate an average of SEK 89 million worth of rebalancing flows. Daily rebalancing flows are demonstrated to contribute to both end-of-day price pressure and volatility in underlying stocks, with more severe effects during times of high market volatility or crisis. The indirect impact from trading in anticipation of rebalancing accounts for the bulk of the total end-of-day price and volatility impacts in underlying stocks. Using predictions from market microstructure invariance, we calculate implied price impacts for both the Financial Crisis and the Corona Crisis to show that rebalancing flows may have contributed to substantial price declines and extra volatility during these periods. Implied price impact calculations show that large index moves risk causing a 'cascade' reaction of self-sustaining rebalancing, leading to significant price declines.

Keywords:

Leveraged ETFs, Financial Stability, Market Microstructure Invariance, Price Impact, Sweden

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1. Introduction

1.1. Background and Relevance

Leveraged and inverse exchange-traded funds (LETFs) are structurally similar to regular exchange-traded funds (ETFs) but promise a return multiple of daily index returns. Leveraged ETFs promise a return multiple of an underlying index greater than one, whereas inverse ETFs promise an index return multiple smaller than one. As of January 2018, assets invested in leveraged and inverse ETFs and ETPs (exchange-traded products) amount to \$87.3 billion (Reuters, 2018). For the Swedish LETF market, which is the geographic scope of this study, this figure is SEK 2.2 billion, as of May 12, 2020 (XACT, 2020).

The complexity of this asset class and its aggressive growth have spurred substantial academic inquiry into their market stability impact. The main characteristics subject to inquiry are twofold. Firstly, positive feedback trading takes place when LETFs conduct their daily rebalancing trading. Thus, if underlying asset returns are positive, LETFs must buy more of the underlying asset to maintain a constant stock-to-cash ratio. Correspondingly, if underlying asset returns are negative, LETFs sell. As a result, one may anticipate that the dynamic portfolio rebalancing of LETFs amplifies return and volatility through non-fundamental positive feedback trading. Additionally, it is likely that these rebalancing trades have a comparatively larger effect due to their price-insensitive nature and condensed time-interval of execution. Secondly, due to the mathematically predictable nature of these trades, there is likely to be a degree of equivalently non-fundamental anticipatory trading undertaken by arbitrageurs prior to LETF rebalancing trades.

The aggregate effects of anticipatory trades and rebalancing trades may 1) prove debilitating to markets in turbulence, 2) reduce market confidence and 3) lead to market wide sell-offs or rallies as a consequence of the price impact derived from rebalancing trades (Tuzun, 2014).

Rebalancing trades occur daily shortly before market close in order to mitigate tracking error and to maintain a fixed stock-to-cash ratio. The rebalancing trades of LETFs in Sweden are executed in the futures market, where index arbitrageurs propagate the effects of LETFs' futures trading to the underlying basket securities. The size of rebalancing flows depends on the LETF's total net assets (TNA), the target return multiple, and the daily index movement. With approximately SEK 2.2 billion in total net assets for OMXS30-tracking LETFs as of April 20, 2020, a 1 % index move causes LETFs to originate SEK 89 million worth of rebalancing flows, as of April 20, 2020. Tuzun (2014) implements a model of price impact developed in Kyle and Obizhaeva (2012) to show that the effects of rebalancing flows are notable, especially during the Financial Crisis 2008-2009, when markets experienced substantial daily volatility and a large subsequent need for rebalancing. Thus, in times of high volatility,

lower liquidity and overall market turbulence, it is plausible that rebalancing trades could pose elevated risks to market stability.

Daily Swedish LETF rebalancing volumes are relatively small fractions of the daily volumes of stocks in the OMXS30 but constitute a substantially larger fraction of the volume during the last hour of trading (see Figures II-IV). For a 1 % index move, rebalancing flows are associated with a price impact in the average stock in the OMXS30 index of 4.0 basis points and 11.8 basis points of extra end-of-day volatility. During the Corona Crisis, a 1 % index move is associated with an increased price impact amounting to 8.9 basis points in an average stock in the OMXS30. This result is driven primarily by increased volatility in OMXS30 index constituents. Furthermore, we establish support for the prevalence of anticipatory trading and show that, for a 1 % index move, anticipatory trading is associated with a price impact of 5.5 basis points in an average stock in the OMXS30. Equivalently, trading in anticipation of expected rebalancing flows in response to a 1 % index move increases end-of-day volatility with 16.4 basis points for an average stock in the OMXS30. Thus, the bulk of end-of-day price and volatility impacts across stocks seems to be derived from the indirect effect of rebalancing (anticipatory trading), rather than the direct effect of daily rebalancing trades.

These estimates may be understated if prices for underlying assets already reflect the anticipated future rebalancing impact, or if traders take the opposite sides of rebalancing trades (Tuzun, 2014). Thus, to estimate the total effect of rebalancing trades, we employ the price impact equation from Kyle & Obizhaeva (2012) to determine the total index level price impact of rebalancing flows implied from index moves of 1 %.

Implied price impacts from rebalancing in excess of 1 % per 1 % index is a critical level for market stability since this creates a risk for a ‘cascade reaction’ where rebalancing trade volumes increase exponentially and become self-sustaining. By computing historical implied price impacts, we show that this level was exceeded for several months during the Financial Crisis, and for several weeks during the contemporary Corona Crisis. Therefore, we provide support for the idea that LETFs may have contributed notably to market turbulence during the contemporary Corona Crisis and the Financial Crisis.

Finally, we conclude that implied price impacts from rebalancing trades are insignificant in ‘normal’ markets but may prove destabilising to markets experiencing high volatility or crisis.

1.2. Research Question

Our research seeks to determine the end-of-day volatility and price pressure impact from rebalancing trades and anticipatory trading in the Swedish market. Furthermore, we aim to outline the market stability effect of LETFs during volatile times, by explicitly analysing the OMXS30 index development between February 24, 2020 and April 20,

2020, defined as the Corona Crisis. Our overarching research question can therefore be defined as follows:

What is the market stability impact of LETFs' rebalancing trades in the Swedish stock market?

1.3. Contribution

We consider the contribution of this thesis to be fourfold. Firstly, many studies have covered LETFs' market stability impact on the US market (Tuzun, T., 2014; Trainor, W.J., 2010; Cheng, M. & Madhavan, A., 2009; Bai, Q., Bond, S. A. & Hath, B., 2012; P. Shum et. al., 2016; Ivanov, I.T., Lenkey, S.L., 2018), however, to the best of our knowledge, there has been no comparable study conducted which covers the Swedish market. Secondly, through precise knowledge of LETF trade execution, along with the analytically appealing Swedish LETF market setting, we hope to arrive at more robust conclusions by excluding confounding factors present in previous papers covering the US market. These conditions also allow us to test for the prevalence and impact of anticipatory trading, which has remained an untested hypothesis in previous literature. Thirdly, since the LETFs' methods of trade execution and general market characteristics vary over time, it is valuable to provide continuous and contemporary research on the market stability impact of LETFs. Fourthly, by explicitly analysing the contemporary Corona Crisis, we provide up-to-date results on the impact of rebalancing flows during times of high market volatility.

1.4. Scope

This study has been limited to the Swedish market and the OMXS30 index due to the nature of our data, the unique institutional setting implied by XACT's monopoly position and our precise knowledge of rebalancing trade execution in the Swedish market. Our intraday index and stock-level analysis is conducted on a sample period ranging from September 19, 2019 until April 20, 2020. The remainder of the analysis conducted is based on a sample period ranging from April 18, 2005, which was the first trading day of the first Swedish LETF, through April 20, 2020.

1.5. Disposition

The remainder of this thesis is divided into nine sections. *Section 2* presents the theories used in the paper along with previous literature pertinent to the thesis. *Section 3* presents the data used along with the adjustments and calculations undertaken to generate key variables. *Section 4* introduces LETFs and presents descriptive statistics and figures for rebalancing, along with empirical tests on the performance of Swedish LETFs. In *section 5*, we present and report the results from various empirical tests on end-of-day price and volatility impacts in underlying stocks. Subsequently, we describe the empirical tests

used to show the prevalence of anticipatory trading and mean-reversion of price impacts and report the results from these regressions. *Section 6* limits the sample period for several regressions conducted in section 5 to explicitly analyse the market stability impact of rebalancing during the contemporary Corona Crisis. *Section 7* uses the implied price impact equation yielded from market microstructure invariance to determine the historical index level price impacts of rebalancing flows. Following this, we deliver concluding remarks in *section 8*.

2. Theory, Practice and Literature Review

This section provides the general mathematical framework behind LETF rebalancing along with some comments regarding its important aspects. Subsequently, we provide firm-specific institutional details of Swedish LETFs managed by XACT, along with some specifications of the nature of rebalancing trade execution. After that, we describe the mechanism that propagates demand shocks from the index futures market to the underlying index constituents. Following this, we present a literature review of the previous academic papers pertinent to this thesis, along with a conceptual description of market microstructure invariance used to compute historical implied price impacts of rebalancing flows.

2.1. Rebalancing Dynamics of Leveraged and Inverse ETFs

In order to deliver the promised multiples of index returns, LETFs are required to rebalance in response to price movements in the underlying asset. The daily rebalancing ensures that LETFs maintain constant stock-to-cash ratios at the end of each day. A simple asset allocation setting can illustrate the mechanics of LETF rebalancing, and the asset allocation problem faced by fund managers can be written in the following way: (Cheng and Madhavan, 2009):

$$W_t = S_t + B_t$$

A certain fraction, δ , is invested of fund managers' net assets, W_t , into the risky asset (the underlying asset tracked by the LETF) S_t , and the rest $(1 - \delta)$ is invested into the bond market.

$$W_t = \delta \times W_t + (1 - \delta) \times W_t$$

$$\text{Where } \delta \times W_t = S_t \text{ and } (1 - \delta) \times W_t = B_t$$

The fraction δ is determined based on the respective LETFs' prospectuses. In our setting, XACT BULL has $\delta = 1.5$ and promises a multiple of 1.5 the index return, while XACT BULL 2 has $\delta = 2$ and is promising a multiple of 2 the index return. The same logic applies for the inverse ETFs, where XACT BEAR has $\delta = -1.5$ and XACT BEAR 2 has $\delta = -2$. Given a change in the index by r , the investment in the index becomes $\delta (1 + r) W_t$, assuming that the interest rate equals zero.

$$W_t(1 + \delta \times r) = \delta(1 + r)W_t + (1 - \delta)W_t$$

$$\text{Where } W_t(1 + \delta \times r) = W_{t+1} \text{ and } \delta(1 + r)W_t = S'_t$$

$\delta \times W_{t+1}$ should be invested in the index to maintain a constant stock-to-cash ratio, promising an index return multiple of δ .

$$S_{t+1} = \delta \times W_{t+1} = \delta \times (1 + \delta \times r) \times W_t$$

The amount of rebalancing can be calculated as follows:

$$S_{t+1} - S'_t = r \times \delta \times (\delta - 1)W_t \quad (1)$$

As noted in Tuzun (2014), an important aspect to this is that when $\delta \notin [0,1]$, the rebalancing value and r have the same sign, meaning that both leveraged and inverse ETFs rebalance in the same directions as the return in the indices they track. Since they do not cancel each other out, the positive feedback mechanism present in both categories of LETFs is crucial to their potentially adverse market impact.

Moreover, the rebalancing is a function of the index change, and not the level of the index, making LETF rebalancing sensitive only to the target index changes and not the target index level, as described in Tuzun (2014).

To exemplify, when the OMXS30 declined 10.57 % during the Black Thursday Crash of March 12, 2020, XACT BULL 2 rebalancing amounted to 21.14 % of its total net assets ($10.57 \% \times 2 \times (2 - 1)$). XACT BEAR 2, on the other hand, rebalanced 63.42 % of its total net assets ($10.57 \% \times -2 \times (-2 - 1)$).

2.2. XACT's LETFs' Institutional Details

When XACT offered XACT BULL in February 10, 2005, it was the first LETF in the world. Since then, LETFs have grown globally and can now be considered a mainstream asset class. However, XACT has remained the sole provider of LETFs covering the Swedish market. This monopoly position, along with the precise knowledge of trade execution we have obtained, allow us to make more accurate estimates of market impact, and to distinguish between the indirect price impact from anticipatory trading and the direct impact from rebalancing trades executed during end-of-day.

Generally, LETFs dynamically rebalance at or near market close, in order to mitigate tracking error (Avellaneda & Zhang, 2010). In this paper, we refer to the time interval of daily rebalancing trade execution for Swedish LETFs as $[T_s, T_c]$ where T_s denotes the starting time of daily rebalancing and T_c denotes market close.

The total rebalancing flow which occurs over the time interval is referred to as total rebalancing flow (T-FLOW) and may impact prices of underlying securities in the OMXS30. Furthermore, seeing as these flows are predictable, we can also expect some degree of anticipatory trading. Therefore, by the time rebalancing takes place, prices may to some degree already reflect the expected rebalancing (Tuzun, 2014). Thus, T_s is

later used to distinguish between underlying stock returns resulting from anticipatory trading and returns resulting directly from mechanical end-of-day rebalancing trades.

Another demand shock predicated by structure of ETFs is the process of creation and redemption of ETF shares (C&R), associated with excess demand or supply for the ETF. Among other purposes, such as tax efficiency, this mediator of arbitrage activity is also the process that ensures that LETF shares trade at, or close to, their net asset value (NAV) (Ben-David et. al, 2017). Gastineau (2001) describes the dynamics of the process through which ETF shares are created and redeemed as follows:

For ETF share creation, an authorised participant (AP), such as an institutional trading desk, purchases a basket of the ETF's underlying securities. The AP then delivers these underlying assets to the ETF and receives an ETF share. The AP then sells this ETF share in the market. The procedure for ETF share redemption works inversely: The AP purchases an ETF share in the market and exchanges this share with the ETF for a basket of the ETF's underlying securities. Subsequently, the AP sells the basket of underlying securities in the market. These C&R trades are typically done in blocks of 50,000 shares of the ETF and are executed at market close.

While, at first, it may appear that C&R-flows do not act as a price discovery channel for its basket securities, T. Ernst (2019) shows that informed traders trade both the ETF and the underlying asset concurrently. The degree of concurrent trading is largest for stocks with a high index weight and large stock-specific informational asymmetry. Thus, due to the purely non-fundamental nature of rebalancing flows, as opposed to C&R-flows, the conclusions regarding the market stability implications of C&R trades vis-a-vis rebalancing trades ought not to be equated. C&R-flow is therefore only to be included as a control variable.

The SEK value of C&R-flow for fund j at time t is calculated using the following formula, as described in Staer (2014):

$$C\&R - FLOW_{j,t} = (Shrout_{j,t-1} - Shrout_{j,t}) \times NAV_{j,t} \quad (2)$$

Where $C\&R - FLOW_{j,t}$ is creation and redemption flow in SEK, $Shrout_{j,t}$ is the number of shares outstanding at close and $NAV_{j,t}$ is the Net Asset Value at close. Therefore, this equation quantifies the exact monetary equivalent of the shares exchanged with the AP at market close.

2.3. Futures Market Shock Propagation

LETFs do not trade directly in the stock market to rebalance their portfolios. Rather, LETFs are believed to employ derivatives such as futures or swaps to quench their need for rebalancing. Despite not trading directly in the underlying securities, index arbitrageurs transfer price shocks from the futures market to the stock market, ensuring

that prices for index futures accurately reflect the prices of its' basket securities. If LETFs rebalance their portfolios using the swap market, counterparties are likely to hedge their positions in the futures market or the spot market, whereby they will leave a corresponding imprint on the stock market (Cheng and Madhavan, 2009; Bai, Q, et.al, 2012).

Through arbitrage trading, the ETF asset class is believed to have facilitated shock propagation between the futures market and the equity market during the Flash Crash of May 6, 2010 (Ben-David et. al, 2012).

2.4. Literature Review

2.4.1. Leveraged and Inverse ETFs

Since the first ETF was offered in 1993, they have grown tremendously, garnering substantial scientific inquiry into their institutional details as well as their impact on global markets. Globally, the wider ETF asset class has grown from one fund in 1993 to 102 funds by 2002 and nearly 1000 by the end of 2009. As of recently, there are at least 5000 ETFs trading globally. In late 2019, their assets under management (AUM) surpassed \$6 trillion (ETF Trends, 2019; Investopedia 2019).

Their leveraged and inverse equivalents, LETFs, were introduced in 2005 – and their total AUM has grown quickly since then, amounting to \$87.3 billion as of January 2018 (Reuters, 2018). Their structure and characteristics, particularly the need for frequent rebalancing in the same direction as the index movement, has spurred significant inquiry into their impact on market stability.

For the US market, Tuzun (2014) finds that a 1 % index move in broad stock market indices induce LETFs to originate \$1.04 billion worth of stock in rebalancing flows. Further, Tuzun demonstrates that end-of-day rebalancing trades contributes to price reactions and extra volatility in underlying stocks, propagated through arbitrage channels. Tuzun relates the positive-feedback trading nature of rebalancing flows to the dynamic replication of portfolio insurance, believed to have contributed to the stock market crash in 1987, and concludes that while rebalancing volumes are smaller, they could pose a threat to market stability in times of high market volatility.

Utilising the price impact equation yielded from market microstructure invariance, Tuzun finds that a 1 % change in price for financial stocks after the crash of Lehman Brothers in 2008 yields rebalancing flows sufficient to elicit a 1.5 % price move, deemed to be a critical level for market stability.

Sampling the S&P 500 index returns and aggregate LETF rebalancing volume, Cheng & Madhavan (2009) find that aggregate rebalancing demand causes price pressure on end-of-day S&P 500 index returns. Similarly, Bai, Q et. al. (2012) find evidence for both end-of-day price pressure and extra volatility, impacted by the rebalancing flows of 6 LETFs covering 63 real estate sector stocks. Contrary to these findings, Trainor (2010) is unable to establish a significant link between LETFs and volatility.

The positive feedback trading of LETFs makes them structurally similar to the portfolio insurance strategies prevalent in the late ‘80s. The Brady Report (1988) argues that selling by portfolio insurers accounted for a substantial portion of the volume during the crash of October 19, 1987. Furthermore, the report notes that certain “aggressive-oriented institutions” traded in advance of this rebalancing, as is believed to occur in advance of LETF rebalancing flows. The aggregate effect of this trading is suggested to have contributed significantly to the 29 % 1987 crash in the S&P 500. Kyle & Obizhaeva (2012) use predictions from market microstructure invariance to examine the aggregate price impact of selling by portfolio insurers and find a predicted price decline of 16.77%.

2.4.2 Market Microstructure Invariance

Market microstructure invariance is a set of hypotheses developed by Albert S. Kyle & Anna A. Obizhaeva. The hypotheses posit that the distributions of risk transfers (“bets”) and transaction costs are constant across assets when measured in units of business time, rather than calendar time, (Kyle & Obizhaeva, 2016). A *bet* is a transaction of an asset of a certain quantity that is intended to lead to a gain for the investor based on the beliefs this person holds regarding the future price of that asset. Market microstructure invariance relies on the intuition that these bets take place in business time. The rate at which business time passes, “market velocity”, is the rate at which new bets enter the market, as opposed to our common notion of time, referred to as “calendar time”. For assets with high turnover, bets arrive quickly, and business time passes quickly. For assets with low turnover, bets arrive slowly, and business time passes slowly.

Market microstructure characteristics such as bet size, market impact and bid-ask spreads vary across assets and time. However, market microstructure invariance hypothesises that these characteristics are constant “microstructure invariants” when measured in business time. Kyle & Obizhaeva go on to formulate two invariance principles.

- The distribution of the dollar risk transferred by a bet is invariant in units of business time.
- The expected dollar transaction cost of executing a bet is an invariant function of the size of the bet when the bet’s size is measured as the dollar risk it transfers in units of business time.

The invariance principles above are tested by analysing a data set of more than 400,000 portfolio transition orders over the period 2001-2005. Examining this data, Kyle & Obizhaeva establish support for the two invariance hypotheses postulated.

The invariance relationships presented above place strong restrictions on the size distribution of bets and their market impact as a function of dollar trading volume and volatility, allowing Kyle & Obizhaeva (2012) to derive a log-linear price impact equation which quantifies the impact of selling X shares of an asset. The implied price impact is given by:

$$\ln\left(1 + \frac{\Delta P(X)}{P}\right) = \frac{\lambda}{10^4} \times \left(\frac{P \times V}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma}{0.02}\right)^{4/3} \times \frac{X}{(0.01)V} \quad (3)$$

where P denotes price, λ is a constant calibrated from portfolio transition orders, V is daily trading volume and σ is the daily volatility of the asset. In a later section, we implement a modified version of this formula to determine the implied price impact of LETF rebalancing flows on the OMXS30 index.¹

¹ See Kyle & Obizhaeva (2016) for the mathematical foundations of market microstructure invariance. See Kyle and Obizhaeva (2012) for the derivation of the implied price impact equation.

3. Data

This section provides a general data description, followed by the different datasets we have used. After this, we report the most noteworthy data processing and data adjustments conducted. Relevant summary statistics are provided in section 4.

3.1. Data Sources

Due to being constrained to a relatively short sample period and the smaller relative size of the Swedish LETF market, we anticipate that results may have lower significance than those demonstrated in previous papers. Thus, we have resolved to minimise any unnecessary approximations by utilising the most granular data available. Information about Swedish LETFs has been obtained from XACT, the sole issuer of LETFs in Sweden. XACT has provided data on NAV, price and daily shares outstanding for all LETFs. The dataset also includes intraday and daily index return series and index volumes for NASDAQ OMXS30, which have been obtained from Börsdata and Bloomberg. Membership history of NASDAQ OMXS30 has been provided by Thomson Reuters. Intraday data for the OMXS30's index constituents has been obtained from the Swedish House of Finance (SHoF). Daily historical USD/SEK and EUR/SEK exchange-rates have been obtained from Yahoo Finance. GDP-deflator data has been obtained from the U.S. Bureau of Economic Analysis. The sample ranges from April 18, 2005, when the first data about shares outstanding is available for XACT BULL and XACT BEAR, to April 20, 2020.

3.2. Data Processing

LETF rebalancing flow was calculated for the respective LETFs using equation 1 and aggregated to receive total rebalancing flow. The returns used in the calculation of rebalancing flows were based on close-to-close returns of the OMXS30GI, which is the target index of all XACT LETFs. Creation and redemption flows (C&R-flow) are calculated using equation 2. Using intraday data from SHoF with one-minute resolution, along with index membership history data, we reconfigure the OMXS30 index on a minute-by-minute basis.² We utilise the composed OMXS30 index to calculate index weights for the underlying assets. The measures of rebalancing flow and C&R-flow are distributed pro rata to the OMXS30's index constituents. The index weights used in the data analysis are based on the index weight at market close. Various intraday return, volatility and volume measures are calculated across stocks. April 9, 2020 is excluded from the sample due to incomplete data. Nordea Bank ABP is included in the index weight calculations but is excluded from the regression samples due to data availability issues subsequent to their re-domiciliation to Finland.

² Unreported results show that the reconfigured OMXS30 index very closely tracks the actual return series of the OMXS30 as reported by Bloomberg.

4. LETF Rebalancing and the Market

After the launch of LETFs in Sweden in 2005, their total net assets grew rapidly up until 2010, when total Swedish LETF TNAs peaked beyond 6 billion SEK. Since then, the TNAs of Swedish LETFs have gradually decreased, as demonstrated in Figure I. This fall in TNA may be partially explained by the rise of literature on the long-term divergence of LETF returns from promised return multiples. For example, Avellaneda & Zhang (2010) demonstrate that the path-dependence of LETF returns may render them unsuitable for buy-and-hold investors.

Table I shows the target multiples, the TNA, the rebalancing flows in response to a 1 % index move for each of the LETFs tracking the OMXS30 and the ratio of the rebalancing flows in response to a 1 % index move over TNA. The largest LETF in Sweden is XACT BEAR 2 with SEK 995 million in total assets, as of April 20, 2020. An index move of 1 % generates SEK 60 million of XACT BEAR 2 rebalancing flows, which equals 6.0 % of TNA. XACT BULL 2 with SEK 520 million in total assets generates SEK 10 million of rebalancing flows in response to a 1 % index move, which equals 2.0 % of TNA. XACT BEAR, with SEK 459 million in total assets, generates rebalancing flows in response to a 1 % index change of 17 million SEK, which equals 3.75 % of TNA. The smallest LETF in the Swedish market, XACT BULL, with total assets of SEK 287 million, generates SEK 2 million of rebalancing flows in response to a 1 % index move, which equals 0.75% of total TNA. In summation, the total rebalancing flow in response to a 1 % index move across all LETFs sums to SEK 89 million, as of April 20, 2020.

The ratios of the rebalancing flows in response to a 1 % index move to TNA demonstrate that LETFs with higher target multiples rebalance a larger fraction of their TNAs in response to index moves. The ratios also show that Inverse ETFs rebalance a larger fraction of their TNA than Leveraged ETFs for equal absolute values of their respective return multiples. Comparatively, Inverse ETFs may therefore pose a more significant threat against markets.

4.1 The Performance of Swedish LETFs vis-a-vis OMXS30

LETF performance can be assessed based on how well they reduce tracking error. The following regression has been used to assess the performance of LETFs (Tuzun, 2014), seeing as they promise multiples of OMXS30 returns:

$$R_{j,t} = \alpha + \beta(\delta_j \times R_t^{OMXS30}) + \epsilon_{j,t} \quad (4)$$

Where $R_{j,t}$ is the return of fund j at time t , δ_j is the return multiple of fund j and R_t^{OMXS30} is the return of the OMXS30GI index at time t . Since XACT funds reinvest dividends, we use the OMXS30GI, which includes dividends. In Table II, regression coefficients and adjusted R-squared are reported for all LETFs (with return multiples, δ , listed in Table I).

For all LETFs, α is not significantly different from zero and ranges from -0.55 basis points for XACT BULL to 0.55 basis points for XACT BEAR. Similarly, β is close to 1 for all LETFs, ranging from 0.976 for XACT BEAR 2 to 0.982 for XACT BEAR. However, β is significantly smaller than 1 on the 1 % level. Moreover, the adjusted R-squared figures for the respective regressions are high, indicating that, on average, Swedish LETFs are successful in achieving the target multiples of daily returns.

4.2. Rebalancing Flow Descriptives

In Table III, summary statistics of daily averages for the market (OMXS30) is presented for two time intervals, the full intraday sample and the Corona Crisis. For the full intraday sample period, the average daily volatility is 1.9 % and the average daily volume is approximately SEK 167 billion, of which SEK 91 million can be attributed to rebalancing flows. During this period, the average daily C&R-flow amounted to SEK 9 million. The average stock in the OMXS30 index had a daily volume of SEK 5.6 billion. The Corona Crisis sample shows an average daily volatility of 3.4 %, an average daily volume of SEK 235 billion, of which SEK 224 million can be attributed to rebalancing flows. The average daily C&R-flow for this period amounted to SEK 19 million. The average stock in the OMXS30 index during the Corona Crisis had a daily volume of SEK 7.8 billion SEK.

Figure II, III and IV show rebalancing flows as a fraction of daily volume, last hour of trading volume and for the volume in the interval $[T_s, T_c]$ for an average stock in the OMXS30 index. The rebalancing flows have been calculated as the aggregate absolute value of all LETFs' rebalancing flows. The volumes for each time period are the sum of all OMXS30 index constituents' volumes during each time frame. The red vertical line in the figures marks the start of the Corona Crisis.

The daily rebalancing ratio before the Corona Crisis exceeded 2 % only five days in the full intraday sample period. During the Corona Crisis, the daily rebalancing ratio exceeded 3 % six days and amounted to roughly 5 % at its peak, which occurred on Black Thursday, March 12, 2020. The figures displaying rebalancing ratios based on volumes during the last hour of trading as well as volumes during the rebalancing interval show that, during these time intervals, the respective rebalancing ratios peak at roughly 14 % and 22 %, also occurring on Black Thursday, during the Corona Crisis. The higher rebalancing ratios during days of extra high volatility suggest that the increase in daily volume associated with crashes do not offset the increase in rebalancing flows. As demonstrated, rebalancing flows as a fraction of end-of-day volumes can therefore be substantial.

Notably, for 15 days during the intraday sample, AstraZeneca experienced implied rebalancing flows exceeding the actual volumes during the interval of rebalancing trade execution. More interestingly, 12 of these 15 days occurred during the Corona Crisis. On the day with the largest price decline, Black Thursday March 12, 2020, AstraZeneca's

pro rata share of total rebalancing amounted to 192.2 % of volume.³ One explanation for these occurrences is that shock propagation through index arbitrage does not work perfectly. In that case, end-of-day price and volatility impacts in underlying assets may understate the actual aggregate market impact of rebalancing flows. However, this requires further research.

Figure V shows rebalancing flows in response to a 1 % index change along with daily volatility for the previous 20 days. The rebalancing flows are hypothetical in the sense that they are estimated as a response to a 1 % index change. Figure VI illustrates how total net assets have varied over time for the Inverse ETFs (BEAR and BEAR 2). By comparing the two graphs, it can be noted that the LETF rebalancing flows in response to a 1 % index move tends to co-move with Inverse ETF's TNA and daily volatility.

Thus, market volatility appears to be associated with an increased demand for Inverse ETFs, suggesting that investors use Inverse ETFs either as a hedging instrument or as an instrument for speculation on market declines. Either way, TNAs for Inverse ETFs appear to increase in times of crisis.

³ Results from unreported analyses.

5. Flow-Induced Market Impact of LETFs

This section provides a description of the set of regression models used to estimate the end-of-day price impact and volatility induced by rebalancing flows. Subsequently, we test for next-day price reversals and the price impact from anticipatory trading. Results from the regressions are commented on throughout the respective sections.

5.1. End-of-day Price Impact in Underlying Securities

As discussed in previous sections, rebalancing flows may exacerbate price moves and cause excess volatility in underlying stocks prior to market close. Through the monopoly position of XACT in the Swedish LETF market, as well as knowledge of the time interval of rebalancing trade execution, we aim to distinguish between the price impact from rebalancing trades and the price impact resulting from anticipatory trading. We implement the following regression specification, as in Tuzun (2014), but include scaled creation and redemption flows as an additional control variable, in order to estimate the price impact of T-flow in basket securities between T_s and market close (c):

$$\frac{\Delta \log(P_{i,c})}{\sigma_i} = \beta_0 + \beta_1 \times \frac{T-FLow_i}{ADV_i} + \beta_2 \times \frac{CR-FLow_i}{ADV_i} + \beta_3 \times \frac{\Delta \log(P_{i,T_s})}{\sigma_i} + \epsilon_i \quad (5)$$

The dependent variable is constructed as a ratio of two variables where $\Delta \log(P_{i,c})$ is the return of stock i from T_s to market close and σ_i is the daily volatility. Similarly, explanatory variables are constructed as ratios of two variables. $T-FLow_i$ is the pro rata share of total LETF rebalancing flow in SEK attributable to stock i based on its' end-of-day index weight. $T-FLow_i$ is scaled by ADV_i which is the average daily volume of stock i . $CR-FLow_i$ is included as a control variable to capture the price effect of creation and redemption flows, which is similarly scaled by ADV_i . The second control variable, $\Delta \log(P_{i,T_s})$ is the log-return of stock i from previous close to T_s scaled by its daily volatility.⁴ Scaling return variables with daily volatility adjusts for heteroskedasticity across stocks (Obizhaeva, 2012). All variables included in the regression are winsorized on the 1 % and 99 % level to remove the effect of outliers. Standard errors are clustered by date. Table IV reports the results of the regression for the full intraday sample period, September 19, 2019 through April 20, 2020. The coefficient on rebalancing flow is significant, and of similar magnitude, when control variables are included. The inclusion of control variables slightly improves the R-squared of the regression model. The coefficient on rebalancing flow is positive and significant on the 1 % level. The coefficient on rebalancing flow can be interpreted as the expected return as a % of daily volatility of stock i in response to rebalancing flows equivalent to 1 % of the average daily volume of stock i . For an average stock in the OMXS30 index, rebalancing flows equivalent to 1 % of the daily volume is associated with an expected

⁴ Average daily volumes and daily volatilities are calculated on a rolling basis over a 20-day period.

price impact of 1.499 % of the stock’s daily volatility. The coefficient on C&R-flows is also positive but insignificant.

Rebalancing flows in response to a 1 % index move amounts to an average of 1 % of daily volume of the average stock in the OMXS30 with a daily volatility of 2.7 % for an average stock. The price reaction for the average stock in the OMXS30 in response to a 1 % index move is therefore 4.0 basis points ($1.499 \times 1 \% \times 2.7 \%$), all else equal. Thus, results suggest that LETF rebalancing causes end-of-day price impacts in OMXS30 index constituents.

The results presented on rebalancing trades’ end-of-day price impact do not differ markedly in kind or size from what has been presented in previous papers. Thus, the Swedish stock market appears to experience end-of-day price impacts from rebalancing trades, similar to the findings from the US market, described in Tuzun (2014),

In Table V results are reported for the regression above, but for positive returns, ‘market ups’, and negative returns, ‘market downs’, separately. Variables are winsorized on the 1 % and 99 % level. Standard errors are clustered by date. The coefficients are significant and estimated at 1.837 for ‘market ups’ and 1.108 for ‘market downs’. These results suggest that there is a stronger market reaction to positive LETF flows compared to negative LETF flows. Tuzun (2014) finds the same pattern in the US market and argues that a higher coefficient for ‘market ups’ may be predicated by short-sale constraints partially inhibiting investors’ abilities to trade opposite LETF rebalancing flows for days with positive market movements.

5.2. End-of-day Volatility Impact in Underlying Securities

As with the price impact established in the section above, LETF rebalancing may also exacerbate end-of-day volatility. To test this, we implement the regression specification for volatility in Tuzun (2014) and allow for C&R-flows as an additional control variable:

$$\left[\frac{\Delta \log(P_{i,c})}{\sigma_i}\right]^2 = \beta_0 + \beta_1 \times \left|\frac{T-FLOW_i}{ADV_i}\right| + \beta_2 \times \left|\frac{CR-FLOW_i}{ADV_i}\right| + \beta_3 \times \left[\frac{\Delta \log(P_{i,T_s})}{\sigma_i}\right]^2 + \epsilon_i \quad (6)$$

The left side of the regression, $\left[\frac{\Delta \log(P_{i,c})}{\sigma_i}\right]^2$ is the squared return between T_s and market close for stock i scaled by its daily variance. $\left|\frac{T-FLOW_i}{ADV_i}\right|$ is the absolute value of total rebalancing flow scaled by the average daily volume attributable to stock i . $\left|\frac{CR-FLOW_i}{ADV_i}\right|$ is the absolute value of creation and redemption flows attributable to stock i , similarly scaled by the average daily volume. $\left[\frac{\Delta \log(P_{i,T_s})}{\sigma_i}\right]^2$ is the squared return for stock i between previous close and T_s scaled by its daily return variance.⁵ All variables included in the regression are winsorized on the 1 % and 99 % level to remove the effect of outliers.

⁵ Same as above.

Standard errors are clustered by date. Table VI reports the results of the regression. The coefficient on the absolute value of rebalancing flow is positive and significant and increases and remains significant upon the inclusion of the control variables. The R-squared increases when control variables are included.

The coefficient on rebalancing flow can be interpreted as the change in return variance in % of the daily variance in response to rebalancing flows equal to 1 % of the daily volume. Thus, for an average stock in the OMXS30, rebalancing flows equal to 1 % of daily volume increase its return variance by 0.19 % of the stock's daily variance.

Average rebalancing flows in response to an index move of 1 % amount to 1 % of daily volume. The average stock in the index has a daily volatility of 2.7 %. The end-of-day volatility impact of a 1 % index move is therefore 11.8 basis points ($\sqrt{0.192 \times 1 \% \times 2.7 \%}$), all else equal. Therefore, the results suggest that LETF rebalancing trades contribute to end-of-day volatility in the OMXS30. Furthermore, the coefficient on $\left[\frac{\Delta \log(P_{i,T_s})}{\sigma_i}\right]^2$ is positive and significant, suggesting that high intraday volatility continues through end-of-day.

5.3. Anticipatory Trading

5.3.1. End-of-Day Price Impact

Due to the mathematically predictable nature of rebalancing flows, we expect some degree of anticipatory trading. However, previous papers covering markets with several LETF providers, coupled with uncertainty regarding rebalancing trade execution, have been unable to distinguish between the price impact derived from anticipatory trading and the direct impact of rebalancing flows. We partially disentangle these effects by differentiating between returns prior to and after T_s , the inception of rebalancing flows. Since anticipatory trading is likely to occur during the time period of rebalancing as well, we do not claim to fully disentangle the effects, but rather to show the prevalence of anticipatory trading and its size relative to the direct price impact estimated earlier.

As in previous regressions, we calculate rebalancing flow based on close-to-close returns. However, this may not to be the basis for decisions by anticipatory traders, unless these traders include the expectation of their own price impact on security prices. Furthermore, it is plausible that anticipatory traders adjust their positions as index returns develop. Therefore, basing expected rebalancing flows on close-to-close returns seems like a reasonable proxy for actual anticipated rebalancing, on average. Furthermore, this makes comparisons between the results from regression 5 more meaningful. Therefore, we set:

$$E [T - FLOW_i] = T - FLOW_i$$

C&R-flows are excluded from the regression since these are deemed difficult to anticipate. To test for anticipatory trading, we implement the following regression:

$$\frac{\Delta \log(P_{i,16:30-T_s})}{\sigma_i} = \beta_0 + \beta_1 \times \frac{T-FLow_i}{ADV_i} + \beta_2 \times \frac{\Delta \log(P_{i,16:30})}{\sigma_i} + \epsilon_i \quad (7)$$

The dependent variable is the log-return between 16:30 and T_s . The explanatory variables are rebalancing flow and log-return between previous day's market close and 16:30. All variables are scaled as in previous sections and winsorized on the 1 % and 99 % level to remove the effect of outliers. Standard errors are clustered by date. Results from the regression are reported in Table VII. The coefficient on rebalancing flow expected by anticipatory traders is positive and significant upon the inclusion of the control variable. The coefficient on rebalancing flow is about 37 % larger than the coefficient in section 5.1, suggesting that the bulk of the total price impact derived from LETF rebalancing is attributable to anticipatory trading.

The coefficient can be interpreted as the price impact for stock i in % of daily volatility, derived from anticipatory trading in expectation of rebalancing flows equal to 1 % of daily volume. Similar to section 5.1, the price impact of anticipatory trading in an average stock in response to a 1 % index move equals 5.5 basis points ($2.050 \times 1 \% \times 2.7 \%$).

As demonstrated from the coefficients in Table VII, compared to Table IV, it appears that anticipatory trading accounts for the bulk of the net end-of-day price impact. Thus, operating under the assumption that anticipatory traders at least partially liquidate their positions against same-day rebalancing flow, the return and volatility effects of rebalancing may be distributed over a longer period. In turn, this may reduce the respective total impacts, compared to a case where no anticipatory trading takes place, and price-insensitive rebalancing trades are instead executed over a short period.

However, significant anticipatory trading may cause issues related to reverse causality. This may be the case if anticipatory trading (which is based on expected rebalancing flows) determine close-to-close index returns, rather than the other way around. Theoretically, a large triggering index move early in the trading day may therefore cause a significant and prolonged deviation from fundamental index value.

While many previous papers have argued that it is likely that anticipatory trading is prevalent, conclusive results have not been presented – to the best of our knowledge. Our results not only show that it is prevalent, but that it appears to account for the bulk of the aggregate non-fundamental impact induced by LETFs' rebalancing trades. These findings imply that the price impacts of LETF rebalancing is not concentrated to the time interval in which rebalancing trades are executed, but rather that the effects are more widely distributed over a longer time period.

5.3.2 End-of-Day Volatility Impact

To test the volatility impact of anticipatory trading, we utilise equation 6, but calculate squared returns from 16:30 to T_s . As in regression equation 7, C&R-flows are not included since its value ought to be difficult to anticipate. We implement the following regression specification:

$$\left[\frac{\Delta \log(P_{i,16:30-T_s})}{\sigma_i} \right]^2 = \beta_0 + \beta_1 \times \left| \frac{T-FLow_i}{ADV_i} \right| + \beta_2 \times \left[\frac{\Delta \log(P_{i,16:30})}{\sigma_i} \right]^2 + \epsilon_i \quad (8)$$

The left-hand side of the regression, $\left[\frac{\Delta \log(P_{i,16:30-T_s})}{\sigma_i} \right]^2$, is the squared return between 16:30 and T_s for stock i scaled by its daily variance. $\left| \frac{T-FLow_i}{ADV_i} \right|$ is the absolute value of total rebalancing flow expected by anticipatory traders, attributable to stock i , scaled by the average daily volume of stock i . $\left[\frac{\Delta \log(P_{i,16:30})}{\sigma_i} \right]^2$ is the squared return for stock i between previous close and 16:30, scaled by its daily return variance. All variables included in the regression are winsorized on the 1 % and 99 % level to remove the effect of outliers. Standard errors are clustered by date. Table VIII reports the results of the regression.

The coefficient on the absolute value of expected rebalancing flow is positive and significant, both including and excluding control variables. It can be interpreted as the change in return variance in % of the daily variance, caused by anticipatory traders who expect future rebalancing flows equal to 1 % of the daily volume. Thus, for an average stock in the OMXS30, expected rebalancing flows equal to 1 % of daily volume increase its return variance by 1.006 % of the stock's daily variance. Average expected rebalancing flows in response to an index move of 1 % amount to 1 % of daily volume. The average stock in the index has a daily volatility of 2.7 %. The average end-of-day volatility impact from anticipatory trading in response to a 1 % index move is therefore 16.4 basis points ($\sqrt{1.006 \times 1 \% \times 2.7 \%}$) in an average stock in the OMXS30, all else equal.

The volatility impact from anticipatory trading has neither been conclusively presented in previous papers, to the best of our knowledge. Similar to the results from the price impact analyses, the volatility impact of anticipatory trading appears to account for the bulk of the extra end-of-day volatility as well.

5.4 Mean-reversion and Resilient Markets

Fama (1991) describes an efficient capital market as one which completely and correctly reflects all information relevant to determining the prices of securities. Thus, according to the efficient market hypothesis, LETF rebalancing trades ought not to affect security prices, seeing as they do not contain any price-relevant information. On the contrary, we find that LETF rebalancing is associated with end-of-day price and volatility impacts, resulting directly from rebalancing and from trading in anticipation of this expected non-fundamental rebalancing. In a resilient capital market, this non-fundamental price

impact should revert. This is supported by empirical studies such as Serlitis and Rosenberg (2007) who find evidence that prices in the US stock market display anti-persistence (mean-reversion). Similarly, Tuzun (2014) demonstrates next-day price reversals associated with previous day rebalancing flows. On the other hand, Graflund (2000) finds no support for mean-reversion in any of the Nordic Stock markets.

We test mean-reversion over two intervals and implement the following regression specifications, similar to Tuzun (2014):

$$\frac{\Delta \log(P_{i,09:30})}{\sigma_i} = \beta_0 + \beta_1 \times \left(\frac{T-FLOW_i}{ADV_i} \right)_{t-1} + \beta_2 \times \left(\frac{CR-FLOW_i}{ADV_i} \right)_{t-1} + \beta_3 \times \left(\frac{\Delta \log(P_i)}{\sigma_i} \right)_{t-1} + \epsilon_i \quad (9)$$

$$\frac{\Delta \log(P_{i:T_s})}{\sigma_i} = \beta_0 + \beta_1 \times \left(\frac{T-FLOW_i}{ADV_i} \right)_{t-1} + \beta_2 \times \left(\frac{CR-FLOW_i}{ADV_i} \right)_{t-1} + \beta_3 \times \left(\frac{\Delta \log(P_i)}{\sigma_i} \right)_{t-1} + \epsilon_i \quad (10)$$

The regressions differ only in the time-period over which mean-reversion is calculated. In equation 9, this is the return from previous close to 09:30 and in equation 10 the return is calculated from previous close to T_s . Explanatory variables are previous day rebalancing flow, C&R-flow and return, scaled as in previous regressions. All variables are winsorized on the 1 % and 99 % level to remove the effect of outliers. Standard errors are clustered by date. Regression results are reported for regressions (9) and (10) in Table IX and X, respectively. With and without control variables, neither regression displays significance, contrary to the findings in Tuzun (2014). However, the coefficients on both rebalancing flow and C&R-flow are negative for returns until 09:30. The lack of significance is likely a result of the short sample period, however, it may indulge the idea that the Swedish stock market is not as efficient as the US stock market. Further analysis is required to make confident statements regarding either proposition.

6. Case Study: The ‘Corona Crisis’

In this section, we analyse the impact of rebalancing trading during the ongoing Corona Crisis, which has caused significant turmoil across global financial markets for several months. As in section 5, we provide price impact and volatility impact regressions, but limit the data range to the Corona Crisis to demonstrate the differential impact of LETF rebalancing flows during times of high market volatility.

6.1. Definitions and Descriptives

The time period for the Corona Crisis is defined as the date range during which the daily OMXS30 index volatility calculated over 10 days exceeds 1.5 %. The resulting date range stretches from February 24, 2020 to the end of our sample, April 20, 2020. The sample data range is very short, suggesting that we may obtain less significant results. Summary statistics of daily averages compared to the full intraday sample are presented in Table III.

6.2. End-of-day Price Impact in Underlying Securities

To test the price impact, we implement regression equation 5 but restrict our sample period to the Corona Crisis. All variables are winsorized on the 1 % and 99 % level to remove the effect of outliers. Standard errors are clustered by date. Results from the regression are reported in Table XI. The coefficient on rebalancing flow is positive and significant with and without control variables. Both the R-squared and the coefficient increases when controls are included. As in 5.1, the coefficient on rebalancing flow can be interpreted as the expected return as a % of daily volatility of stock i in response to rebalancing flows equivalent to 1 % of the average daily volume of stock i . For an average stock in the OMXS30 index, rebalancing flows equivalent to 1 % of the daily volume is associated with an expected price impact of 2.529 % of the stock’s daily volatility during the Corona Crisis.

Rebalancing flows in response to a 1 % index move amount to an average of 0.8 % of daily volume of the average stock in the OMXS30 with a daily volatility of 4.4 %, during the Corona Crisis. The price reaction for the average stock in the OMXS30 in response to a 1 % index move is therefore 8.9 basis points ($2.529 \times 0.8 \% \times 4.4 \%$). Thus, results suggest that LETF rebalancing flows are associated with more substantial price impacts during periods of high market volatility.

Notably, despite the shorter sample period, the coefficient on LETF rebalancing flow has a higher level of significance during the Corona Crisis than for the full intraday sample period, suggesting that price impacts are more pronounced during times of crises.⁶

⁶ The differential impact between the coefficient on rebalancing flow in the regression of anticipatory trading compared to the direct rebalancing flow impact is of similar magnitude as what was reported in section 5.4 (roughly 37 %) during the Corona Crisis. Thus, anticipatory trading seems to account for the bulk of the price impact in volatile times as well.

6.3. End-of-day Volatility Impact in Underlying Securities

To test end-of-day volatility, we use regression equation 6 but limit our sample to the date range of the Corona Crisis. All variables are winsorized on the 1 % and 99 % level to remove the effect of outliers. Standard errors are clustered by date. Regression results are reported in Table XII. The coefficient on the absolute value of rebalancing flow is not significant, both including and excluding control variables. This is likely due to a combination of the short sample period and significant end-of-day market movements and uncertainty during the period.

7. Price Impact and Market Microstructure Invariance

In this section, we will report the price impacts implied by market microstructure invariance from the first trading day of the first Swedish LETF, until April 20, 2020. Furthermore, we provide support for the idea that price impacts from rebalancing flows may have contributed to market volatility and mispricing during times of crises.

7.1. Implied Price Impact

Due to the difficulty in determining the point in time when prices begin to incorporate the expectation of future rebalancing and the plausible presence of opposite bets, we can resort to the implied price impact equation as described in section 2.4.2 to infer the total implied price impact from rebalancing flows.

Kyle et. al (2011b) demonstrate the internal consistency and predictions of market microstructure invariance by analysing media coverage in relation to stock trading activity. They show that both the trading game and the information flow conform to the same mathematical properties of business time.

Furthermore, Kyle and Obizhaeva (2012) utilise the implied price impact equation to yield the price impact predicted by market microstructure invariance for five stock market crashes. They find that the prediction accuracy of the implied price impact model is high when compared to the observed price declines.

Due to integration and shock transfer across stocks in an index, as argued in Kyle and Obizhaeva (2012) and Tuzun (2014), we analyse the implied price impacts on an index level rather than ditto for individual stocks. However, as should be noted, if the futures market and the market for underlying assets are closely integrated, the measure of ADV using OMXS30 constituents' aggregate volume may be understated, and the price impact may be overstated. Naturally, this logic can be extended, and one could argue for integration across international stock markets. However, due to plausibly imperfect integration across markets, we estimate ADV as in Tuzun (2014).

We analyse the implied price impact by utilising equation 3. We convert SEK volume into USD volume by using a daily USD/SEK rate and adjust for inflation by incorporating an annual GDP-deflator, as in Kyle and Obizhaeva (2012). Equation 3 therefore becomes:

$$\ln\left(1 + \frac{\Delta P_t}{P_{t-1}}\right) = \frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t} \quad (11)$$

Solving for $\frac{\Delta P_t}{P_{t-1}}$ yields:

$$\frac{\Delta P_t}{P_{t-1}} = \exp\left(\frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t}\right) - 1 \quad (12)$$

where λ is a constant calibrated from portfolio transition orders and is estimated at 5.78 (Kyle and Obizhaeva, 2011a), ADV_t is the average daily volume of OMXS30, FX_t is the

daily USD/SEK exchange-rate, DEF_t is the annual GDP-deflator for the United States, σ_t is the daily volatility of the OMXS30 and $Flow_t$ is the hypothetical total daily rebalancing flow on an index level in response to a 1 % index move.⁷

In table XIII, mean values of the variables used in the implied price impact calculation are presented. As evidenced, daily volatility is substantially higher during the Corona Crisis, compared to the full sample.

The decrease in total LETF assets, coupled with the increase in ADV during the Corona Crisis, results in lower scaled LETF flows during the Corona Crisis. However, the mean value of the daily implied price impact during the Corona Crisis amounts to 95 basis points, which is substantially higher than the mean for the full sample period, which amounts to only 11 basis points.

In Figure VII we show the historical implied price impact of aggregate LETF rebalancing flow since the first trading day of the first LETF until April 20, 2020. Figure VIII shows the implied price impact during a period that encapsulates the Financial Crisis. Figure IX shows the implied price impact during the Corona Crisis. The most severe price impacts appear take place during market turmoil such as the Financial Crisis (2007-08), the European Debt Crisis (2009-12) and the Corona Crisis (2020). For non-crisis periods, price impacts remain close to zero. However, several consecutive days during the Corona Crisis exhibit implied price impacts above 1 % per 1 % index move, which is deemed a critical level for market stability due to the potential for a ‘cascade’ reaction, as described in Tuzun (2014). This critical level is shown with a red horizontal line in the figures VIII and IX, for the Financial Crisis and the contemporary Corona Crisis, respectively. Thus, implied price impact calculations suggest that LETF rebalancing flows may have contributed to the large price swings observed during periods of crisis. These findings rhyme with the results from section 6, which empirically demonstrated more substantial price impacts during periods of high volatility.

Figures X and XI present robustness results for the implied impact computation for the period between the first trading day of the first LETF until April 20, 2020. The figures plot the price impact implied by equation 12 with volatilities and ADVs estimated from the past 40 days and 60 days, respectively. The figures demonstrate the robustness of the implied price impact computation. The implied price impacts spike during the same volatile periods as discussed above but have lower amplitudes and swings due to more smoothed daily volatility and ADV estimates. Regardless of calculation periods for ADV and daily volatility, the implied price impacts remain substantial during times of crises, suggesting that LETF rebalancing may have contributed to substantial price declines during these periods.

⁷ Average daily volumes and daily volatilities are estimated on a rolling basis over the past 20 days.

8. Concluding Remarks

8.1 Conclusion

LETFs rebalance their portfolios daily to achieve their promised return multiples. The positive feedback nature of rebalancing trades induces LETFs to sell when index returns are negative, and to buy when index returns are positive. This daily rebalancing is associated with end-of-day price impacts and extra end-of-day volatility in the OMXS30's index constituents. The price impact is more severe during times of high market volatility.

Expected LETF rebalancing is also associated with both end-of-day price and volatility impacts derived from anticipatory trading. This anticipatory trading seems to account for the bulk of said price and volatility impacts, suggesting that the indirect effects of rebalancing are more severe than the direct effects from rebalancing trades. Furthermore, in certain market conditions, it is plausible that significant anticipatory trading can imply prolonged intraday deviations from fundamental index value, compared to a market which experiences no anticipatory trading.

Since the offering of the first LETF, price impacts predicted by market microstructure invariance have been substantial, primarily during the Financial Crisis, the European Debt Crisis and the Corona Crisis. The rebalancing flow price impact implied from a 1 % index move has exceeded 1 % for several days during each crisis, suggesting that LETF rebalancing and anticipatory trading may jointly constitute a risk for a “cascade” reaction if the price impact from rebalancing trades or anticipatory trading elicits significant further rebalancing. A 1 % implied price impact from a 1 % index move can therefore be considered a critical level for market stability.

Rebalancing flow and daily volatility are the main factors influencing this implied price impact, and their effects are multiplicative. As a result of short bets and hedging using Inverse ETFs, hypothetical rebalancing flow per 1 % index change and daily volatility seem to co-move relatively closely. These factors lead to a volatile historical series of implied price impacts, where the most substantial price impacts occur almost exclusively when markets are experiencing high volatility or crisis. Thus, LETFs appear to have a negligible effect on ‘normal’ markets but may prove harmful to markets in crisis.

In an international context, the size of the Swedish LETF asset class is not alone a cause for alarm. However, significant growth of LETF assets could imply a more acute threat to the stability of the Swedish stock market. In a worst-case scenario, significant future growth in LETF assets, combined with a substantial ‘triggering’ index move, could lead to a substantial price decline that develops into a ‘cascade’ reaction, as described earlier. Under such conditions, we may see a market crash similar in kind to the Black Monday Crash of 1987. Consequently, close attention must be paid to the size and composition of LETF assets to ensure that the asset class does not pose an immediate threat to markets, particularly during periods of high market volatility.

8.2. Directions for Future Research

To obtain more significant results and to establish support for the prevalence of mean-reversion of previous day price impacts, the intraday sample period ought to be extended. Additionally, a more granular and detailed analysis of the drivers and exact characteristics of anticipatory trading is required, in order to define its market impact more accurately.

Furthermore, due to their relative importance in determining the aggregate rebalancing flow, an analysis of the sources and timing of the demand for Inverse ETFs is valuable.

A more detailed analysis of the dynamics of shock propagation by index arbitrageurs would also be required to determine the actual market impact of rebalancing trades, which are executed in the futures market rather than the market for underlying assets.

In conclusion, more investigation is required into the subcomponents that cause rebalancing and its respective impacts in order to more accurately grasp the structural threat of the LETF asset class, particularly during turbulent times.

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Appendix

Table I: Leveraged and Inverse ETFs

	Target Multiple	Total Net Assets (SEKm)	Rebalancing Flow r=1 %, SEKm	Ratio of Rebalancing Flow r=1 % / Total Net Assets
XACT BEAR	-1.5	459	17	3.8 %
XACT BEAR 2	-2	995	60	6.0 %
XACT BULL	1.5	287	2	0.7 %
XACT BULL 2	2	520	10	2.0 %
XACT Total		2 261	89	4.0 %

This table presents descriptives on all LETFs on the Swedish market. It presents the LETFs' target multiples of daily index returns, total net assets in million SEK, rebalancing flows in response to a 1 % index move in million SEK and the ratios of rebalancing flows in response to a 1 % index move over total net assets, in SEK. The data presented in this table is as of April 20, 2020.

Table II: Daily Performance of LETFs

XACT Funds	BEAR	BULL	BEAR 2	BULL 2
β	0.982	0.981	0.976	0.977
	(0.003)	(0.003)	(0.003)	(0.003)
α	5.46e-05	-5.51e-05	4.78e-06	-4.13e-05
	(6.28e-05)	(5.65e-05)	(6.76e-05)	(6.58e-05)
# of Observations	3 769	3 769	2 620	2 620
Adjusted R-squared	0.965	0.971	0.979	0.980

Standard errors in parentheses

This table presents the results from the following regression:

$$R_{j,t} = \alpha + \beta(\delta_j \times R_t^{OMXS30}) + \epsilon_{j,t}$$

$R_{j,t}$ is the return of LETF j on a daily basis on day t and R_t^{OMXS30} is the OMXS30 (the target index of LETFs in Sweden) return on day t . δ_j is the target multiple of daily OMXS30 returns of LETF j .

Table III: Index-Level Summary Statistics – Full Intraday Sample vs. Corona Crisis

Daily Averages	Full Intraday Sample	Corona Crisis
Volatility	1.9 %	3.4 %
Volume (SEKm)	167 127	234 952
Rebalancing Flows (SEKm)	91	224
C&R-Flows (SEKm)	9	19

This table reports daily averages of the variables presented. The full intraday time period is defined as September 19, 2019 to April 20, 2020. The Corona Crisis time period has been defined as February 24, 2020 to the last day of the sample, April 20, 2020. The average volatility has been calculated as the standard deviation of index returns (OMXS30) during the respective periods. The average daily volume has been calculated as the average daily volume on index level (OMXS30) for the two periods. The daily average of rebalancing flows has been calculated as the daily average of the sum of all LETFs' rebalancing flows in absolute values for both periods. The daily average of C&R-flows has been calculated as the daily average of the sum of all LETFs' C&R-flows in absolute values for both periods.

Table IV: End-of-Day Price Impact – Full Intraday Sample

End-of-Day Price Impact – Full Intraday Sample		
	T _s to Close	T _s to Close
$\frac{T - FLOW_i}{ADV_i}$	1.116*	1.499***
	(0.631)	(0.545)
$\frac{CR - FLOW_i}{ADV_i}$		2.804
		(3.451)
$\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$		-0.009*
		(0.005)
Constant	0.016**	0.015**
	(0.007)	(0.007)
# of Observations	3 707	3 654
R-squared	0.016	0.023

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the full intraday sample period, defined between September 19, 2019 to April 20, 2020:

$$\frac{\Delta \log(P_{i,c})}{\sigma_i} = \beta_0 + \beta_1 \times \frac{T - FLOW_i}{ADV_i} + \beta_2 \times \frac{CR - FLOW_i}{ADV_i} + \beta_3 \times \frac{\Delta \log(P_{i,T_s})}{\sigma_i} + \varepsilon_i$$

$\frac{\Delta \log(P_{i,c})}{\sigma_i}$ is the return of a stock i between the start of rebalancing to market close (17:30 CET+1), scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\frac{T - FLOW_i}{ADV_i}$ is the share, based on index weight of stock i , of total rebalancing flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\frac{CR - FLOW_i}{ADV_i}$ is the share, based on index weight of stock i , of total C&R-flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . $\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$ is the return of stock i between previous day's close to start of rebalancing scaled by its daily volatility. The standard errors are clustered by date.

Table V: End-of-Day Price Impact – ‘Market Ups’ vs. ‘Market Downs’

End-of-Day Price Impact – Full Intraday Sample		
	Market Ups	Market Downs
$\frac{T - FLOW_i}{ADV_i}$	1.837** (0.832)	1.108** (0.561)
$\frac{CR - FLOW_i}{ADV_i}$	4.421 (2.930)	-2.292 (5.379)
$\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$	-0.032*** (0.007)	-0.035*** (0.011)
Constant	0.044*** (0.008)	-0.025** (0.012)
# of Observations	1 820	1 805
R-squared	0.045	0.036

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for ‘market ups’ and for ‘market downs’. The period is full intraday sample period, defined between September 19, 2019 to April 20, 2020:

$$\frac{\Delta \log(P_{i,c})}{\sigma_i} = \beta_0 + \beta_1 \times \frac{T - FLOW_i}{ADV_i} + \beta_2 \times \frac{CR - FLOW_i}{ADV_i} + \beta_3 \times \frac{\Delta \log(P_{i,T_s})}{\sigma_i} + \varepsilon_i$$

$\frac{\Delta \log(P_{i,c})}{\sigma_i}$ is the return of a stock i between the start of rebalancing to market close (17:30 CET+1), scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days’ returns. $\frac{T - FLOW_i}{ADV_i}$ is the share, based on index weight of stock i , of total rebalancing flows into stock i , scaled by the past 20 days’ average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\frac{CR - FLOW_i}{ADV_i}$ is the share, based on index weight of stock i , of total C&R-flows into stock i , scaled by the past 20 days’ average daily SEK volume for stock i . $\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$ is the return of stock i between previous day’s close to start of rebalancing scaled by its daily volatility. The standard errors are clustered by date.

Table VI: End-of-Day Volatility – Full Intraday Sample

End-of-Day Volatility – Full Intraday Sample		
	T _s to Close	T _s to Close
$\left \frac{T - FLOW_i}{ADV_i} \right $	0.309*** (0.097)	0.192* (0.107)
$\left \frac{CR - FLOW_i}{ADV_i} \right $		0.215 (0.197)
$\left[\frac{\Delta \log(P_{i,T_s})}{\sigma_i} \right]^2$		0.002*** (0.001)
Constant	0.015*** (0.001)	0.014*** (0.001)
# of Observations	3 707	3 654
R-squared	0.019	0.030

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the full intraday sample period, defined between September 19, 2019 to April 20, 2020:

$$\left[\frac{\Delta \log(P_{i,c})}{\sigma_i} \right]^2 = \beta_0 + \beta_1 \times \left| \frac{T - FLOW_i}{ADV_i} \right| + \beta_2 \times \left| \frac{CR - FLOW_i}{ADV_i} \right| + \beta_3 \times \left[\frac{\Delta \log(P_{i,T_s})}{\sigma_i} \right]^2 + \epsilon_i$$

$\frac{\Delta \log(P_{i,c})}{\sigma_i}$ is the return of a stock i between the start of rebalancing to market close (17:30 CET+1), scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\left| \frac{T - FLOW_i}{ADV_i} \right|$ is the absolute value of the share, based on index weight of stock i , of total rebalancing flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\left| \frac{CR - FLOW_i}{ADV_i} \right|$ is the absolute value of the share, based on index weight of stock i , of total C&R-flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . $\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$ is the return of stock i between previous day's close to start of rebalancing scaled by its daily volatility. The standard errors are clustered by date.

Table VII: Price Impact of Anticipatory Trading – Full Intraday Sample
Price Impact of Anticipatory Trading – Full Intraday Sample

	16:30 to T _s	16:30 to T _s
$\frac{T - FLOW_i}{ADV_i}$	1.181	2.050*
	(1.369)	(1.134)
$\frac{\Delta \log(P_{i,16:30})}{\sigma_i}$		-0.023*
		(0.013)
Constant	0.026*	0.025*
	(0.015)	(0.014)
# of Observations	3 504	3 482
R-squared	0.006	0.013

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the full intraday sample period, defined between September 19, 2019 to April 20, 2020:

$$\frac{\Delta \log(P_{i,16:30-T_s})}{\sigma_i} = \beta_0 + \beta_1 \times \frac{T - FLOW_i}{ADV_i} + \beta_2 \times \frac{\Delta \log(P_{i,16:30})}{\sigma_i} + \epsilon_i$$

$\frac{\Delta \log(P_{i,16:30-T_s})}{\sigma_i}$ is the return of a stock i between an hour before market close (16:30 CET+1) to the start of rebalancing, scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\frac{T - FLOW_i}{ADV_i}$ is the share, based on index weight of stock i , of total rebalancing flows, as expected by anticipatory traders, into stock i , scaled by the past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\frac{\Delta \log(P_{i,16:30})}{\sigma_i}$ is the return of stock i between previous day's close to 16.30 CET+1 scaled by its daily volatility. The standard errors are clustered by date.

Table VIII: Volatility Impact from Anticipatory Trading – Full Intraday Sample

Volatility Impact from Anticipatory Trading – Full Intraday Sample		
	16:30 to T _s	16:30 to T _s
$\left \frac{T - FLOW_i}{ADV_i} \right $	1.777*** (0.609)	1.006*** (0.344)
$\left[\frac{\Delta \log(P_{i,16:30})}{\sigma_i} \right]^2$		0.013*** (0.004)
Constant	0.040*** (0.006)	0.033*** (0.006)
# of Observations	3 504	3 482
R-squared	0.043	0.085

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the full intraday sample period, defined between September 19, 2019 to April 20, 2020:

$$\left[\frac{\Delta \log(P_{i,16:30-T_s})}{\sigma_i} \right]^2 = \beta_0 + \beta_1 \times \left| \frac{T - FLOW_i}{ADV_i} \right| + \beta_2 \times \left[\frac{\Delta \log(P_{i,16:30})}{\sigma_i} \right]^2 + \epsilon_i$$

$\frac{\Delta \log(P_{i,16:30-T_s})}{\sigma_i}$ is the return of a stock i between 16:30 CET+1 until start of rebalancing, T_s, scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\left| \frac{T - FLOW_i}{ADV_i} \right|$ is the absolute value of the share, based on index weight of stock i , of total rebalancing flows, as expected by anticipatory traders, into stock i , scaled by the past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\frac{\Delta \log(P_{i,16:30})}{\sigma_i}$ is the return of stock i between previous day's close to 16:30 CET+1 scaled by its daily volatility. The standard errors are clustered by date.

Table IX: Price Reversals, Previous Close to 09:30 – Full Intraday Sample
Price Reversals – Full Intraday Sample

	Previous Close to 09:30	Previous Close to 09:30
$\left(\frac{T - FLOW_i}{ADV_i}\right)_{t-1}$	-0.965 (3.539)	-0.971 (2.930)
$\left(\frac{CR - FLOW_i}{ADV_i}\right)_{t-1}$		-0.582 (21.58)
$\frac{\Delta \log(P_i)}{\sigma_i}$		-0.001 (0.037)
Constant	-0.008 (0.052)	-0.008 (0.053)
# of Observations	3 698	3 669
R-squared	0.000	0.000

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the full intraday sample period, defined between September 19, 2019 to April 20, 2020:

$$\frac{\Delta \log(P_{i,09:30})}{\sigma_i} = \beta_0 + \beta_1 \times \left(\frac{T - FLOW_i}{ADV_i}\right)_{t-1} + \beta_2 \times \left(\frac{CR - FLOW_i}{ADV_i}\right)_{t-1} + \beta_3 \times \left(\frac{\Delta \log(P_i)}{\sigma_i}\right)_{t-1} + \epsilon_i$$

$\frac{\Delta \log(P_{i,09:30})}{\sigma_i}$ is the return of a stock i between the previous day's market close to 09:30 CET+1, scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\left(\frac{T - FLOW_i}{ADV_i}\right)_{t-1}$ is the previous day's share, based on index weight of stock i , of total rebalancing flows into stock i , scaled by previous day's past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\left(\frac{CR - FLOW_i}{ADV_i}\right)_{t-1}$ is the previous day's share, based on index weight of stock i , of total C&R-flows into stock i , scaled by the previous day's past 20 days' average daily SEK volume for stock i . $\left(\frac{\Delta \log(P_i)}{\sigma_i}\right)_{t-1}$ is the previous day's return of stock i , scaled by its daily volatility. The standard errors are clustered by date.

Table X: Price Reversals, Previous Close to Start of Rebalancing – Full Intraday Sample

Price Reversals – Full Intraday Sample		
	Close to T _s	Close to T _s
$\left(\frac{T - FLOW_i}{ADV_i}\right)_{t-1}$	2.270 (4.463)	1.361 (3.872)
$\left(\frac{CR - FLOW_i}{ADV_i}\right)_{t-1}$		-31.11 (28.55)
$\frac{\Delta \log(P_i)}{\sigma_i}$		0.019 (0.043)
Constant	-0.057 (0.071)	-0.050 (0.071)
# of Observations	3 654	3 625
R-squared	0.001	0.004

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the full intraday sample period, defined between September 19, 2019 to April 20, 2020:

$$\frac{\Delta \log(P_{i,:T_s})}{\sigma_i} = \beta_0 + \beta_1 \times \left(\frac{T - FLOW_i}{ADV_i}\right)_{t-1} + \beta_2 \times \left(\frac{CR - FLOW_i}{ADV_i}\right)_{t-1} + \beta_3 \times \left(\frac{\Delta \log(P_i)}{\sigma_i}\right)_{t-1} + \epsilon_i$$

$\frac{\Delta \log(P_{i,:T_s})}{\sigma_i}$ is the return of a stock i between the previous day's market close to start of rebalancing, scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\left(\frac{T - FLOW_i}{ADV_i}\right)_{t-1}$ is the previous day's share, based on index weight of stock i , of total rebalancing flows into stock i , scaled by previous day's past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\left(\frac{CR - FLOW_i}{ADV_i}\right)_{t-1}$ is the previous day's share, based on index weight of stock i , of total C&R-flows into stock i , scaled by the previous day's past 20 days' average daily SEK volume for stock i . $\left(\frac{\Delta \log(P_i)}{\sigma_i}\right)_{t-1}$ is the previous day's return of stock i , scaled by its daily volatility. The standard errors are clustered by date.

Table XI: End-of-Day Price Impact – Corona Crisis

End-of-Day Price Impact – Corona Crisis		
	T _s to Close	T _s to Close
$\frac{T - FLOW_i}{ADV_i}$	1.885*	2.529***
	(1.033)	(0.734)
$\frac{CR - FLOW_i}{ADV_i}$		4.164
		(7.203)
$\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$		-0.017
		(0.014)
Constant	0.015	0.009
	(0.020)	(0.021)
# of Observations	985	956
R-squared	0.074	0.090

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the Corona Crisis period, defined between February 24, 2020 to April 20, 2020:

$$\frac{\Delta \log(P_{i,c})}{\sigma_i} = \beta_0 + \beta_1 \times \frac{T - FLOW_i}{ADV_i} + \beta_2 \times \frac{CR - FLOW_i}{ADV_i} + \beta_3 \times \frac{\Delta \log(P_{i,T_s})}{\sigma_i} + \varepsilon_i$$

$\frac{\Delta \log(P_{i,c})}{\sigma_i}$ is the return of a stock i between the start of rebalancing to market close (17:30 CET+1), scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\frac{T - FLOW_i}{ADV_i}$ is the share, based on index weight of stock i , of total rebalancing flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\frac{CR - FLOW_i}{ADV_i}$ is the share, based on index weight of stock i , of total C&R-flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . $\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$ is the return of stock i between previous day's close to start of rebalancing scaled by its daily volatility. The standard errors are clustered by date.

Table XII: End-of-Day Volatility – Corona Crisis

End-of-Day Volatility – Corona Crisis		
	T _s to Close	T _s to Close
$\left \frac{T - FLOW_i}{ADV_i} \right $	0.203	−0.037
	(0.175)	(0.221)
$\left \frac{CR - FLOW_i}{ADV_i} \right $		1.105
		(1.374)
$\left(\frac{\Delta \log(P_{i,T_s})}{\sigma} \right)^2$		0.002**
		(0.001)
Constant	0.022***	0.022***
	(0.005)	(0.005)
# of Observations	985	956
R-squared	0.009	0.031

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

This table reports the results of the following regression for the Corona Crisis period, defined between February 24, 2020 to April 20, 2020:

$$\left[\frac{\Delta \log(P_{i,c})}{\sigma_i} \right]^2 = \beta_0 + \beta_1 \times \left| \frac{T - FLOW_i}{ADV_i} \right| + \beta_2 \times \left| \frac{CR - FLOW_i}{ADV_i} \right| + \beta_3 \times \left[\frac{\Delta \log(P_{i,T_s})}{\sigma_i} \right]^2 + \epsilon_i$$

$\frac{\Delta \log(P_{i,c})}{\sigma_i}$ is the return of a stock i between the start of rebalancing to market close (17:30 CET+1), scaled by its daily volatility, σ_i , which has been calculated as the standard deviation of the past 20 days' returns. $\left| \frac{T - FLOW_i}{ADV_i} \right|$ is the absolute value of the share, based on index weight of stock i , of total rebalancing flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . The rebalancing flows are a function of the daily index returns. $\left| \frac{CR - FLOW_i}{ADV_i} \right|$ is the absolute value of the share, based on index weight of stock i , of total C&R-flows into stock i , scaled by the past 20 days' average daily SEK volume for stock i . $\frac{\Delta \log(P_{i,T_s})}{\sigma_i}$ is the return of stock i between previous day's close to start of rebalancing scaled by its daily volatility. The standard errors are clustered by date.

Table XIII: Descriptives of Implied Price Impact

Variables	2005-2020	Corona Crisis
ADV (in billion SEK)	153	225
Volatility	1.2 %	3.0 %
$\frac{Flow_t}{(0.01)ADV_t}$	0.064	0.040
$\frac{\Delta P_t}{P_{t-1}}$ (in bps)	11.108	94.805

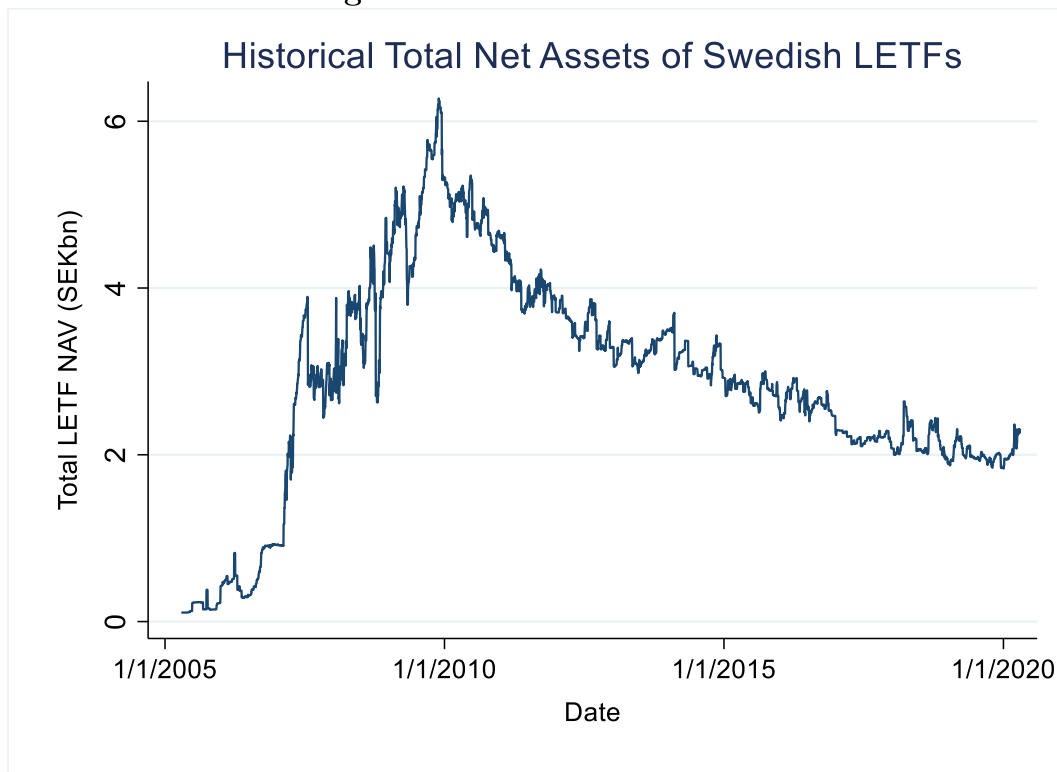
This table reports daily averages of the variables used in the following computation for implied price impact, presented in Kyle and Obizhaeva (2012):

$$\frac{\Delta P_t}{P_{t-1}} = \exp\left(\frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t}\right) - 1$$

ADV is the average daily volume over the past 20 days. The volatility is the average of daily volatilities, calculated as the standard deviation of the past 20 days' index returns.

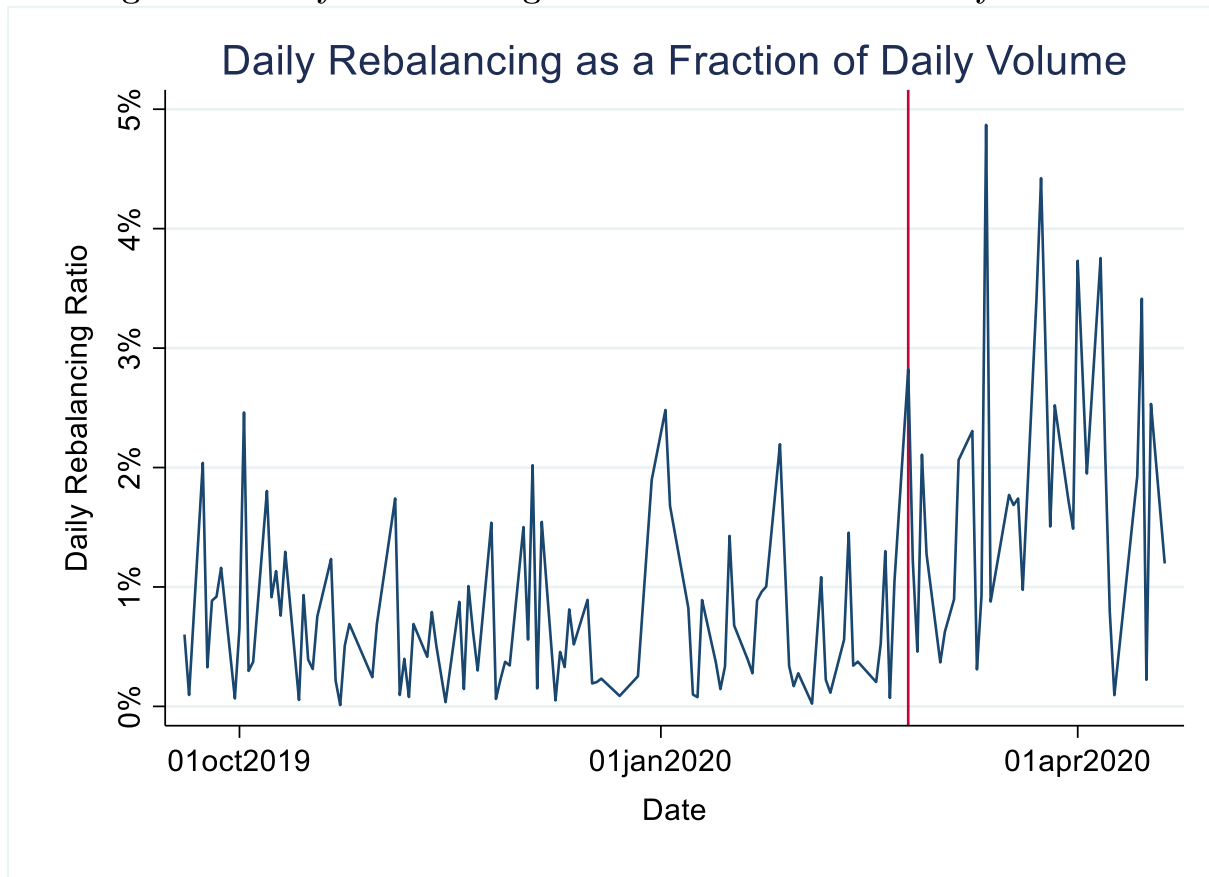
$\frac{Flow_t}{(0.01)ADV_t}$ is the hypothetical total rebalancing flow in response to a one percent index move, scaled by 0.01*ADV. λ is 5.78. FX_t is the daily USD/SEK exchange-rate. DEF_t is the annual GDP deflator for the US economy.

Figure I: Historical Total ETF TNA



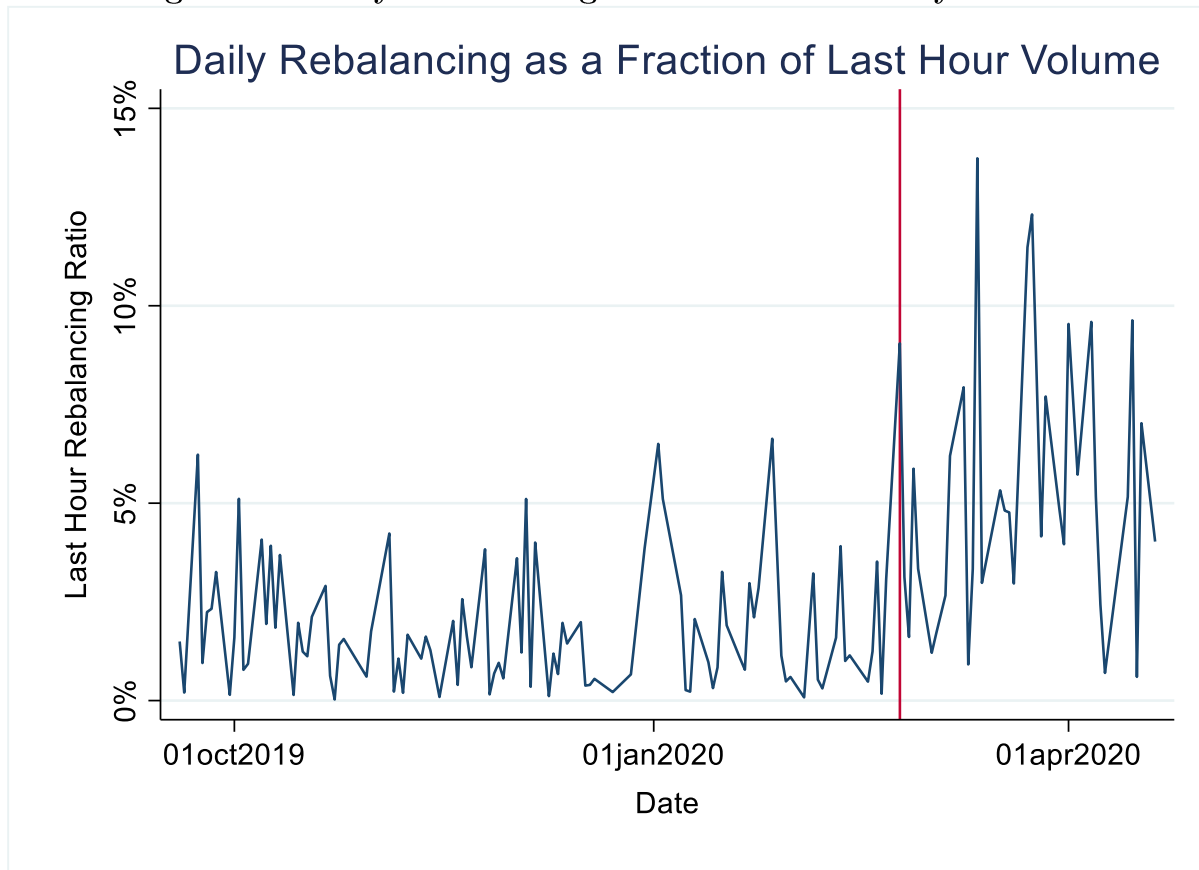
This figure illustrates total TNA of Swedish ETFs between the launch of ETFs in April 2005 through April 2020, in billion SEK.

Figure II: Daily Rebalancing Ratio as a Fraction of Daily Volume



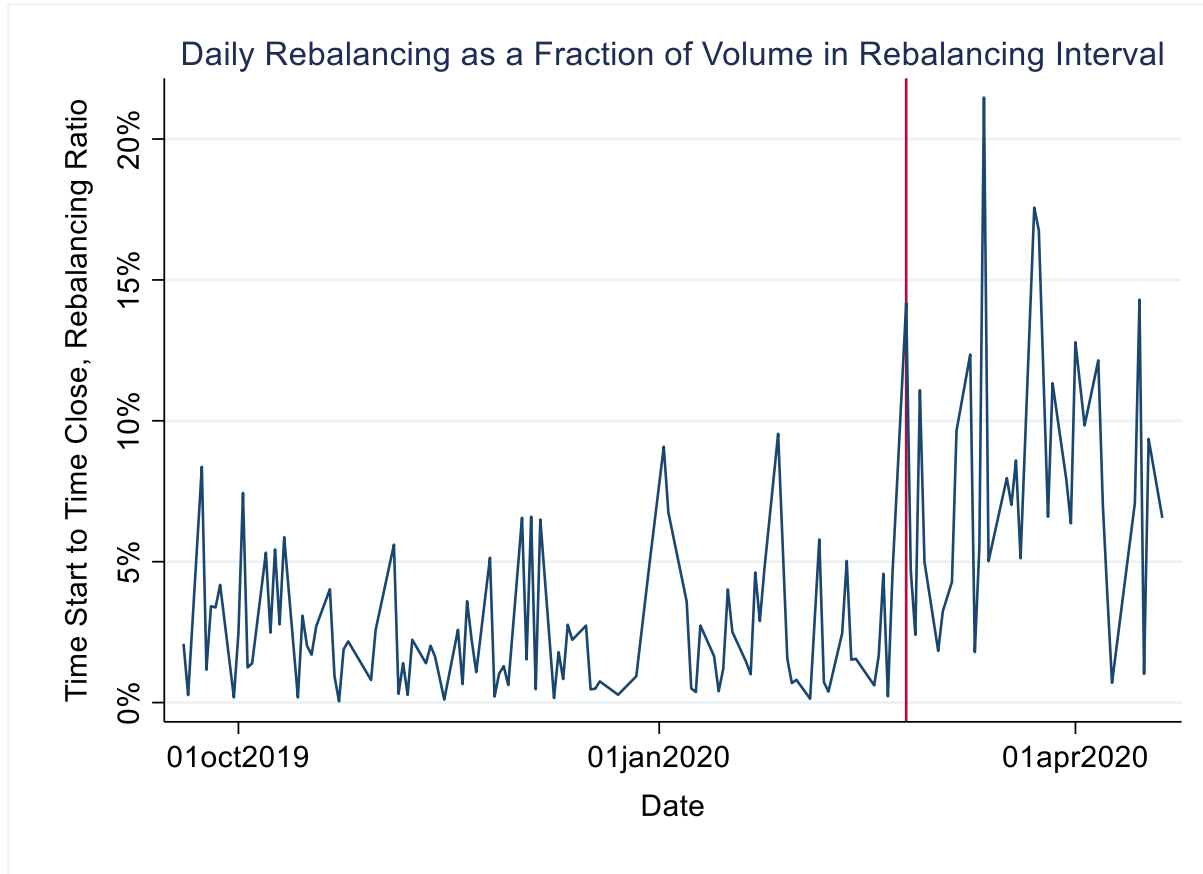
This figure illustrates the share of daily rebalancing as a fraction of daily volume of an average stock in the OMXS30 index. Total LETF flow is the sum of all LETFs' flows. Total LETF flows have been allocated to stocks in the OMXS30 index based on their end-of-day index weights. The volumes of the stocks are daily SEK volumes. The red vertical line marks the start of the Corona Crisis, defined between February 24, 2020 until the end of the sample April 20, 2020.

Figure III: Daily Rebalancing as a Fraction of Daily Volume



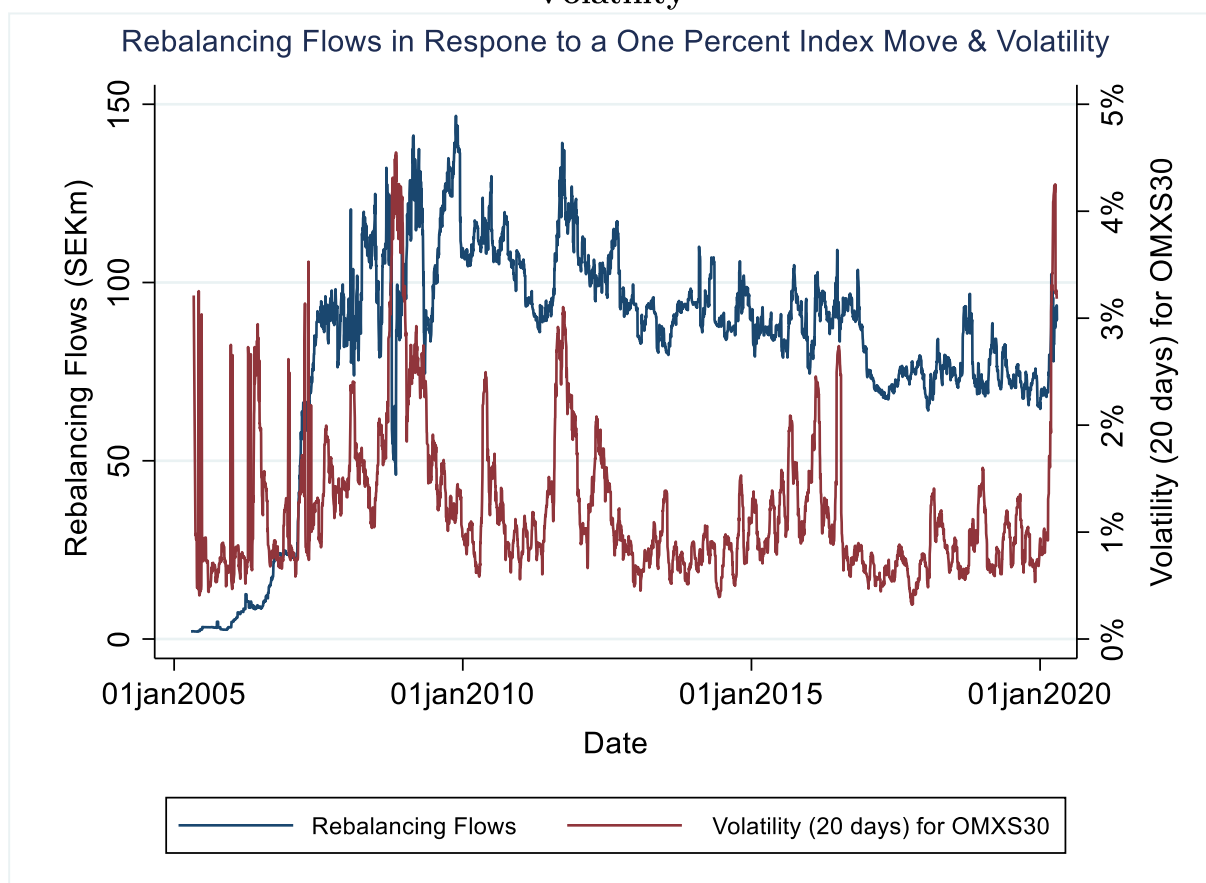
This figure illustrates the share of daily rebalancing as a fraction of last hour volume of an average stock in the OMXS30 index. Total ETF flow is the sum of all ETFs' flows. Total ETF flow has been allocated to stocks in the OMXS30 index based on their index weights. The volumes of stocks are the daily SEK volumes for the last hour of trading (16:30 to 17:30). The red vertical line marks the start of the Corona Crisis, defined as between February 24, 2020, to the end of the sample April 20, 2020.

Figure IV: Daily Rebalancing as a Fraction of Volume in Rebalancing Interval



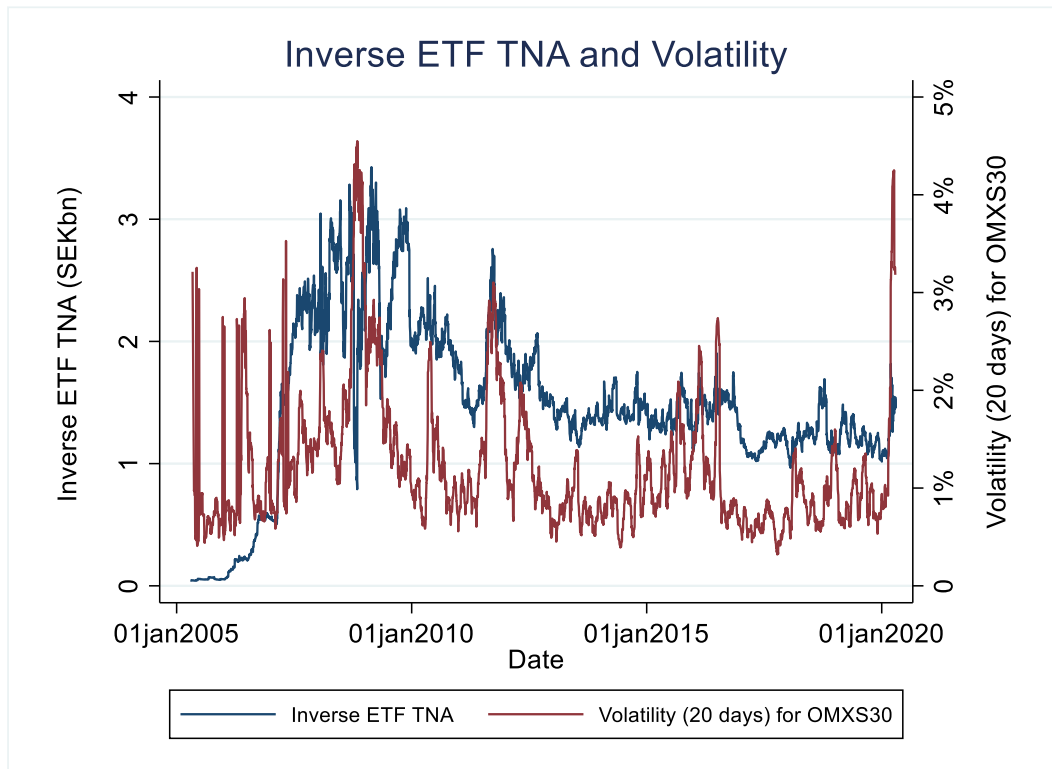
This figure illustrates the share of daily rebalancing as a fraction of volume during the interval of LETF rebalancing (T_s to 17:30) of an average stock in the OMXS30 index. Total LETF flow is the sum of all LETFs' flows. Total LETF flow has been allocated to stocks in the OMXS30 index based on their index weights. The volumes of stocks are the daily SEK volumes between the start of LETF rebalancing, T_s , and market close, 17:30. The red vertical line marks the start of the Corona Crisis, defined as between February 24, 2020, to the end of the sample, April 20, 2020.

Figure V: Rebalancing Flows in Response to a One Percent Index Move & Volatility



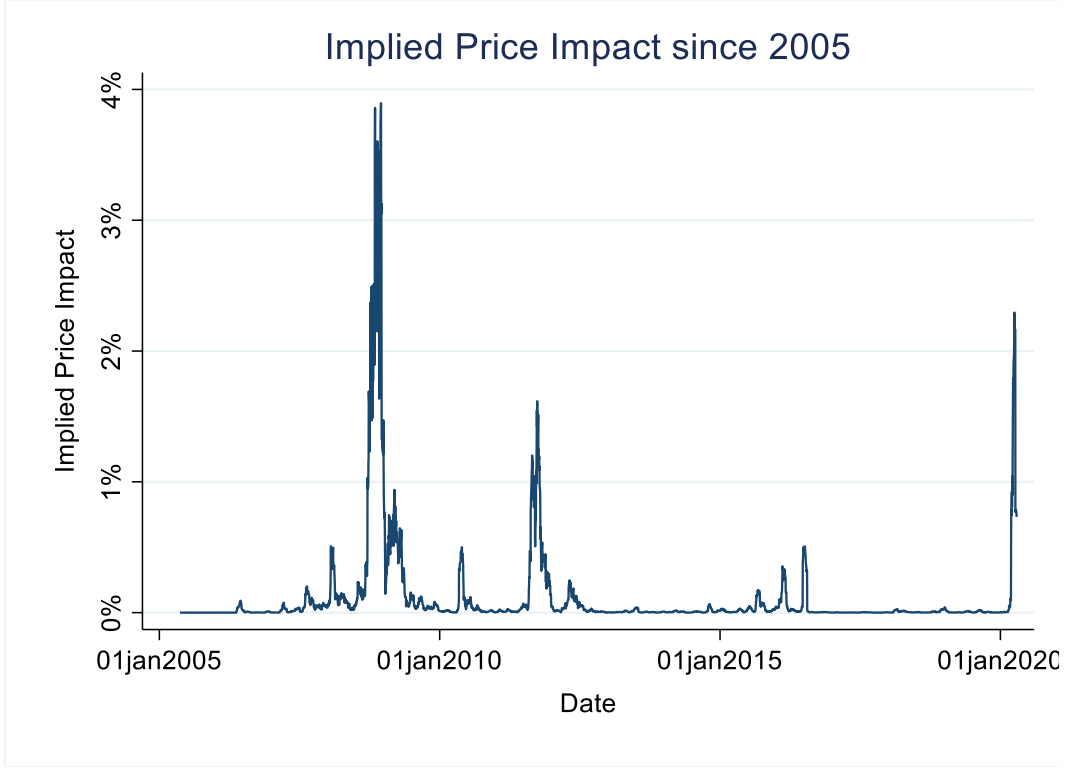
This figure illustrates the theoretical rebalancing flows in response to a one percent index increase and volatility between April 2005 and April 2020. The theoretical rebalancing flows have calculated based on the sum of separately estimated theoretical LETF flows in response to a one percent index increase. Volatility is defined as the daily index level volatility estimated over the previous 20 days' index returns. The left-hand axis is the rebalancing flows in response to a one percent index increase, in million SEK. The right-hand axis is the previous 20 days' volatility in percent.

Figure VI: Inverse ETF TNA and Volatility



This figure illustrates the TNA for all inverse ETFs on the Swedish market and volatility. The TNA for all inverse ETFs has been calculated as the sum of XACT BEAR TNA and XACT BEAR 2 TNA as they constitute all inverse ETFs on the Swedish market. Volatility is estimated from the past 20 days index returns. The left axis is the sum of inverse ETF TNA in billion SEK. The right axis is the previous 20 days' volatility in percent.

Figure VII: Implied Price Impact since 2005

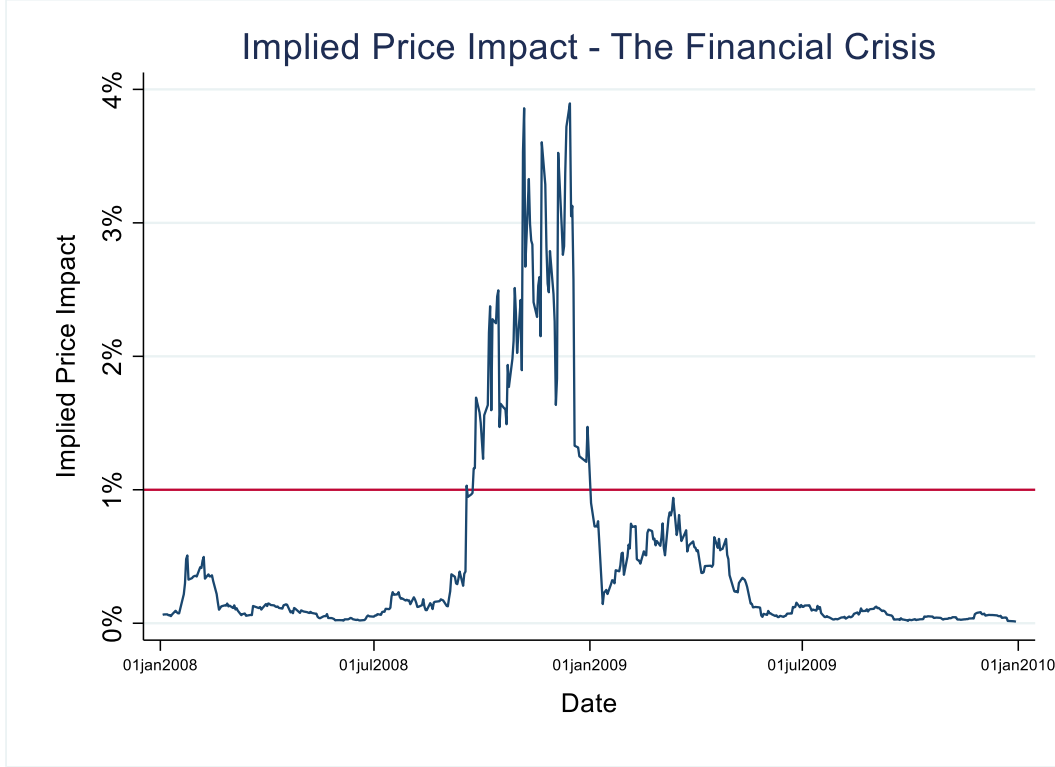


This figure plots estimated price impact, between April 18, 2005 to April 20, 2020, implied by the following computation, presented by Kyle and Obizhaeva (2012):

$$\frac{\Delta P_t}{P_{t-1}} = \exp\left(\frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t}\right) - 1$$

ADV is the average daily volume over the past 20 days. The volatility is estimated as the standard deviation of the past 20 days' index returns. $\frac{Flow_t}{(0.01)ADV_t}$ is the hypothetical total rebalancing flow in response to a one percent index move, scaled by 0.01*ADV. λ is 5.78. FX_t is the daily USD/SEK exchange-rate. DEF_t is the annual GDP-deflator for the US economy.

Figure VIII: Implied Price Impact – The Financial Crisis

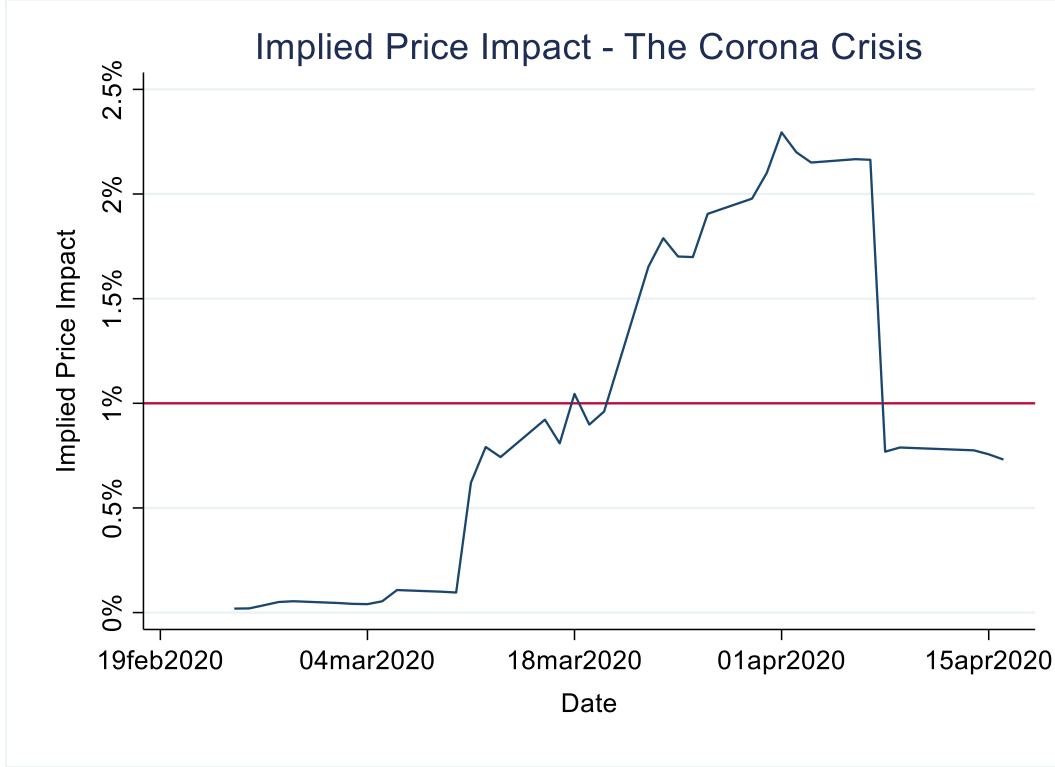


This figure plots estimated price impact, between January 01, 2008 to January 01, 2010, implied by the following computation, presented by Kyle and Obizhaeva (2012):

$$\frac{\Delta P_t}{P_{t-1}} = \exp\left(\frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t}\right) - 1$$

ADV is the average daily volume over the past 20 days. The volatility is estimated as the standard deviation of the past 20 days' index returns. $\frac{Flow_t}{(0.01)ADV_t}$ is the hypothetical total rebalancing flow in response to a one percent index move, scaled by 0.01*ADV. λ is 5.78. FX_t is the daily USD/SEK exchange-rate. DEF_t is the annual GDP deflator for the US economy. The red horizontal line marks the critical level of a 1 % price impact from a 1 % index move.

Figure IX: Implied Price Impact – The Corona Crisis

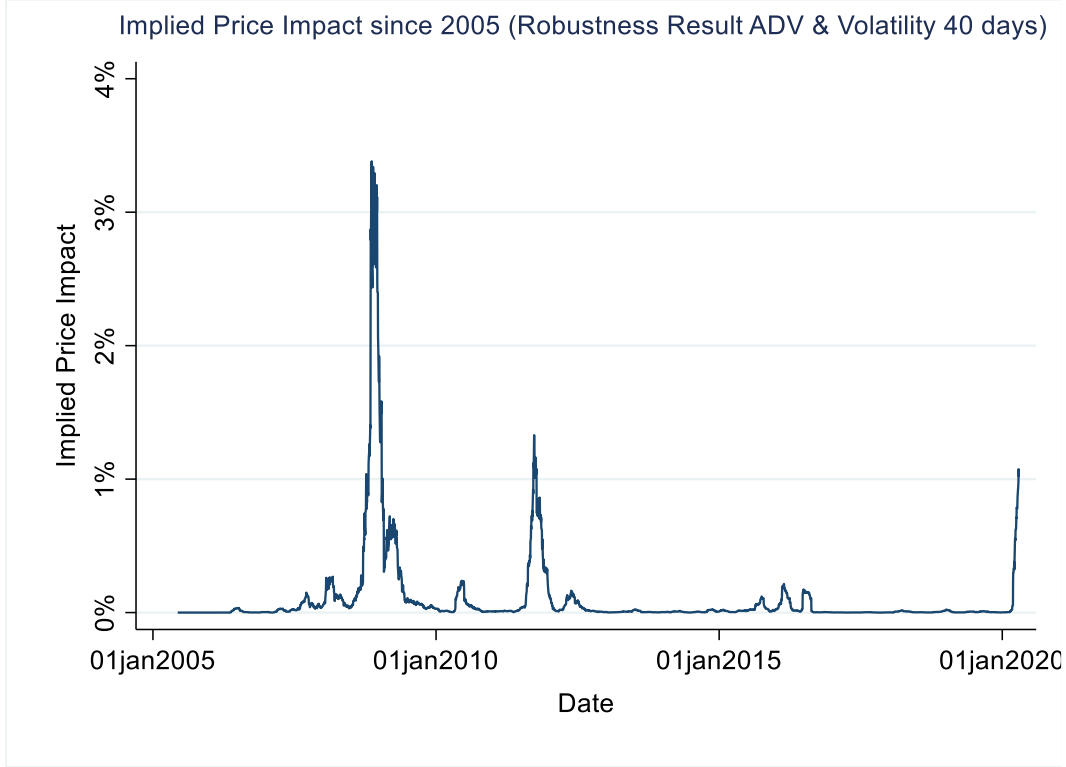


This figure plots estimated price impact, between February 24, 2020 to April 20, 2020, implied by the following computation, presented by Kyle and Obizhaeva (2012):

$$\frac{\Delta P_t}{P_{t-1}} = \exp\left(\frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t}\right) - 1$$

ADV is the average daily volume over the past 20 days. The volatility is estimated as the standard deviation of the past 20 days' index returns. $\frac{Flow_t}{(0.01)ADV_t}$ is the hypothetical total rebalancing flow in response to a one percent index move, scaled by 0.01*ADV. λ is 5.78. FX_t is the daily USD/SEK exchange-rate. DEF_t is the annual GDP deflator for the US economy. The red horizontal line marks the critical level of a 1 % price impact from a 1 % index move.

Figure X: Robustness Result - Implied Price Impact

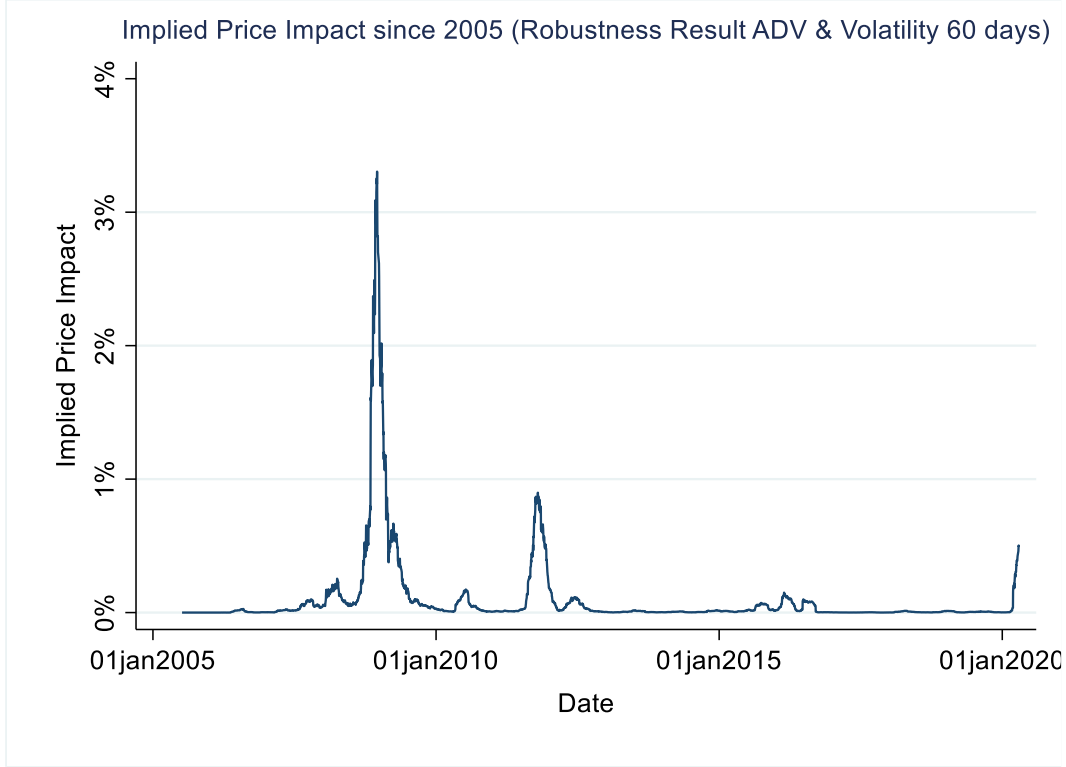


This figure plots a robustness result of estimated price impact, between April 18, 2005 to April 20, 2020, implied by the following computation, presented by Kyle and Obizhaeva (2012):

$$\frac{\Delta P_t}{P_{t-1}} = \exp\left(\frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t}\right) - 1$$

ADV is the average daily volume over the past 40 days. The volatility is estimated as the standard deviation of the past 40 days' index returns. $\frac{Flow_t}{(0.01)ADV_t}$ is the hypothetical total rebalancing flow in response to a one percent index move, scaled by 0.01*ADV. λ is 5.78. FX_t is the daily USD/SEK exchange-rate. DEF_t is the annual GDP deflator for the US economy.

Figure XI: Robustness Result - Implied Price Impact



This figure plots a robustness result of estimated price impact, between April 18, 2005 to April 20, 2020, implied by the following computation, presented by Kyle and Obizhaeva (2012):

$$\frac{\Delta P_t}{P_{t-1}} = \exp\left(\frac{\lambda}{10^4} \times \left(\frac{ADV_t \times (FX_t)^{-1} \times DEF_t}{40 \times 10^6}\right)^{1/3} \times \left(\frac{\sigma_t}{0.02}\right)^{4/3} \times \frac{Flow_t}{(0.01)ADV_t}\right) - 1$$

ADV is the average daily volume over the past 60 days. The volatility is estimated as the standard deviation of the past 60 days' index returns. $\frac{Flow_t}{(0.01)ADV_t}$ is the hypothetical total rebalancing flow in response to a one percent index move, scaled by 0.01*ADV. λ is 5.78. FX_t is the daily USD/SEK exchange-rate. DEF_t is the annual GDP deflator for the US economy.